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CRuSS: Cross-domain Reliability under Shift and Stress for Medical Image Classification

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Abstract:	High in-distribution performance does not guarantee reliable behavior after deployment, yet medical-image classifiers are still commonly ranked using a small set of clean-test discrimination metrics. We introduce CRuSS (Cross-domain Reliability under Shift and Stress), a bounded, transparent reliability score integrating five complementary axes without learned or hand-tuned task weights: clean predictive adequacy, cross-domain performance retention, calibration quality under shift, global corruption resilience, and class-local resilience. Geometric aggregation prevents a severe weakness in one axis from being masked by strength in another. We evaluate CRuSS using 120 clean runs from eight convolutional, transformer, vision-language, and self-supervised model families trained on three four-class brain-MRI datasets, together with 240 no-adaptation source-to-target evaluations across five seeds, 144 controlled-corruption conditions, and 144 class-specific shift conditions. DINOv2-Base achieved the highest CRuSS (98.35; environment-bootstrap 95% CI 97.05–99.39). MobileNetV2 ranked second by mean clean macro-F1 (98.90%) but last by CRuSS (90.30), revealing a clinically relevant rank inversion driven by corruption and class-local fragility. Clean macro-F1 and CRuSS were nearly uncorrelated across models (Spearman ρ=0.048, p=0.911), while random weight perturbation preserved ranking stability (mean Kendall τ=0.871 over 10,000 Dirichlet draws), with DINOv2-Base ranked first throughout. A provenance audit identified substantial transformed-image overlap in a proposed fourth dataset, so those results were excluded from independent-generalization claims. CRuSS is intended as an auditable complement to task-specific metrics, summarizing whether predictive performance survives realistic distributional stress while exposing the source of failure.

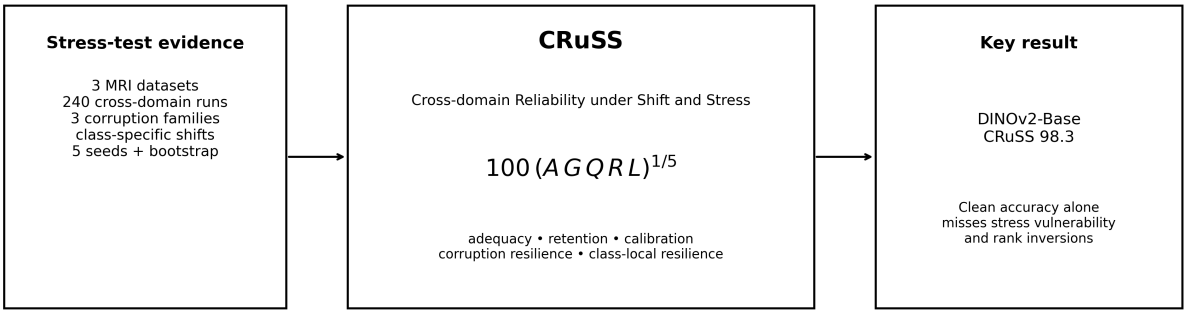
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Graphical Abstract

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Highlights

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- CRuSS integrates five shift-reliability axes without learned task-specific weights.
- Evaluation covers 120 clean runs and 240 no-adaptation cross-dataset tests.
- Clean performance and CRuSS rankings diverge strongly across eight model families.
- Corruption and class-local stress reveal failures hidden by near-saturated clean scores.
- Dataset provenance auditing prevents contaminated external sets from inflating claims.

CRuSS: Cross-domain Reliability under Shift and Stress for Medical Image Classification

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ABSTRACT

High in-distribution performance does not guarantee reliable behavior after deployment, yet medical-image classifiers are still commonly ranked by a small set of clean-test discrimination metrics. We introduce CRuSS (Cross-domain Reliability under Shift and Stress), a bounded, transparent reliability score that integrates five complementary axes without learned or hand-tuned task weights: clean predictive adequacy, cross-domain performance retention, calibration quality under domain shift, global corruption resilience, and class-local resilience. CRuSS uses geometric aggregation so that a severe weakness in one reliability axis cannot be masked by excellence on another. We evaluate the metric using 120 clean runs from eight convolutional, transformer, vision-language, and self-supervised model families trained on three four-class brain-MRI collections, together with 240 no-adaptation source-to-target evaluations across five random seeds, 144 controlled-corruption conditions, and 144 class-specific shift conditions. DINOv2-Base achieved the highest CRuSS (98.35; environment-bootstrap 95% interval 97.05–99.39). Importantly, MobileNetV2 ranked second by mean clean macro-F1 (98.90%) but last by CRuSS (90.30), demonstrating a clinically relevant rank inversion driven by corruption and class-local fragility. Across the eight models, clean macro-F1 and CRuSS were nearly uncorrelated (Spearman $\rho = 0.048$, $p = 0.911$), whereas random weight perturbation preserved the overall ranking well (mean Kendall $\tau = 0.871$ across 10,000 Dirichlet draws), with DINOv2-Base ranked first in every draw. A provenance audit further identified substantial transformed-image overlap in an initially proposed fourth dataset; those results were excluded from independent-generalization claims and retained only as an overlap sensitivity case study. CRuSS is intended as an auditable complement, not a replacement, to task-specific metrics: it summarizes whether strong predictive performance survives realistic distributional stress while exposing the component responsible for failure.

1. Introduction

Deep learning has produced highly accurate medical-image classifiers, but the central deployment question is not simply whether a model performs well on a held-out sample drawn from the same data-generating process. Real deployments change scanners, sites, acquisition protocols, preprocessing, prevalence, image quality, and case mix. Dataset shift therefore creates a gap between benchmark performance and operational reliability [3, 4, 8]. In medical imaging, this gap can be especially consequential because a model may retain a high area under the receiver-operating-characteristic curve while becoming poorly calibrated, disproportionately fragile for one diagnostic class, or unstable under acquisition artifacts.

Recent methodological guidance emphasizes that no single conventional metric is universally adequate for biomedical image analysis. Metrics Reloaded and its companion analysis of metric pitfalls formalize the need to match metrics to the domain interest, data properties, and failure costs [1, 2]. Robustness and reliability, however, extend beyond correctness on a static test set. Strong in-distribution accuracy can coexist with poor external generalization [11, 12]; calibration can degrade under shift [7, 8]; and model selection can be underspecified, with systems that appear equivalent on the development distribution behaving differently under deployment conditions [13].

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The dominant reporting practice nevertheless remains fragmented. Accuracy, balanced accuracy, F1, Matthews correlation coefficient (MCC), AUROC, AUPRC, expected calibration error (ECE), corruption accuracy, and external-dataset performance are typically shown in separate tables. This is necessary for auditability but leaves a practical model-selection problem: how should one summarize reliability when a model is excellent on some dimensions but weak on others? A simple arithmetic average is unsatisfactory because a catastrophic weakness can be compensated by a saturated metric, and arbitrary user-selected weights can make rankings unstable or easy to manipulate.

We address this gap with CRuSS, a Cross-domain Reliability under Shift and Stress score. CRuSS is a bounded geometric aggregation of five explicit components: clean adequacy (A), cross-domain retention (G), calibration quality (Q), global corruption resilience (R), and class-local resilience (L). Each component remains separately reportable. The geometric mean is deliberately bottleneck-sensitive: a model must perform credibly across all dimensions to obtain a high score. No learned coefficients, target-dataset tuning, or post-hoc optimization of weights is used.

The empirical study uses three four-class brain-MRI collections and eight heterogeneous model families: ResNet50 [17], DenseNet121 [18], MobileNetV2 [19], ViT-B/16 [20], DeiT-Small [21], Swin-Tiny [22], CLIP ViT-B/32 [23], and DINOv2-Base [24]. Five random seeds are evaluated. Unlike a conventional architecture paper, the objective is not to propose another classifier; the model families act as diverse probes of an evaluation construct.

The contributions are fourfold. First, we define a transparent reliability score with bounded, monotonic components and a scale-free retention formulation. Second, we validate it under cross-dataset, progressive corruption, calibration, and class-specific stress, exposing rank inversions that clean metrics miss. Third, we quantify score uncertainty and ranking stability through environment-level bootstrap and weight-sensitivity analysis. Fourth, we treat dataset provenance as a prerequisite for reliability claims: a proposed fourth dataset was subjected to exact-hash and deep near-duplicate auditing, and its apparent external performance was not used as independent validation after strong transformed-image overlap was identified.

2. Related work

2.1. Validation metrics in biomedical image analysis

Metric choice is a methodological decision rather than a reporting afterthought. Reinke et al. catalogued failure modes caused by inappropriate metric selection, while Maier-Hein et al. proposed a problem-fingerprint framework for matching metrics to task characteristics [2, 1]. For imbalanced classification, precision–recall analysis can be more informative than ROC analysis [16], and MCC can reveal behavior hidden by raw accuracy [15]. CRuSS does not replace these metrics; instead, it summarizes complementary reliability dimensions after appropriate task-level metrics have been computed.

2.2. Distribution shift and external generalization

Dataset shift is a long-standing problem [3]. Modern benchmarks such as WILDS demonstrate systematic in-distribution to out-of-distribution degradation across real-world tasks [10]. In medical imaging, Zech et al. showed substantial hospital-dependent generalization changes in pneumonia detection [4]. More broadly, repeated benchmark reuse can obscure generalization difficulty [11], and natural distribution shifts may reorder model rankings [12]. These findings motivate a retention-based rather than absolute-only external component.

2.3. Calibration and uncertainty under shift

Modern neural networks can be poorly calibrated even when classification accuracy is high [7]. Bayesian binning and related approaches target probability calibration [14], but Ovadia et al. demonstrated that uncertainty quality can deteriorate under dataset shift [8]. CRuSS therefore includes shifted ECE as an explicit reliability dimension rather than assuming that discrimination and confidence quality move together.

2.4. Corruption robustness and local failure

Common-corruption benchmarks expose sensitivity to blur, noise, and intensity changes that clean tests do not capture [9]. However, global corruptions can still conceal class-local vulnerability. A medical classifier may be robust on average while one diagnosis degrades sharply. We therefore distinguish global corruption resilience from class-local resilience, making the latter a first-class component instead of relying only on macro averages.

2.5. From metric selection to deployment reliability

Metrics Reloaded provides a principled framework for choosing task-appropriate validation metrics, but it intentionally does not collapse every aspect of model quality into a universal scalar [1, 2]. CRuSS addresses a narrower and complementary question: once a set of task-valid measurements has been selected, can the persistence of model quality across multiple deployment-relevant stresses be summarized without hiding a severe failure mode? The distinction is important. CRuSS is not a substitute for problem-aware metric selection; it is an aggregation layer over pre-specified reliability axes whose individual values remain visible.

This deployment-oriented framing is motivated by evidence that clinically relevant operating behavior can drift even when conventional discrimination remains strong. Roschewitz et al. showed that acquisition changes can alter calibrated operating points in medical-image classifiers despite retained ROC-AUC, emphasizing that discrimination alone is insufficient for post-deployment monitoring [30]. Similarly, large-scale self-supervised representation learning has been associated with improved cross-domain transfer in diagnostic imaging, but such gains remain heterogeneous across datasets and tasks [31]. Domain-adaptation studies in biomedical imaging further underscore that scanner, site, and protocol changes induce systematic distribution mismatch rather than random test noise [32]. These observations motivate a score that explicitly separates cross-domain retention, calibration, and corruption behavior instead of treating them as interchangeable manifestations of "accuracy."

2.6. Why a new aggregate is needed

A reliability aggregate is useful only if it adds information beyond a familiar metric. Three design failures are particularly easy to introduce. First, an arithmetic average of several near-saturated metrics can reward redundancy rather than reliability. Second, unconstrained positive and negative terms can make interpretation scale-dependent. Third, fitted weights can overfit the very benchmark used to claim superiority. CRuSS was therefore designed around bounded components, ratio-based retention where appropriate, equal log-space weighting, and explicit sensitivity analysis. The resulting score is intended to behave like a conjunctive index: a high value means that no measured reliability axis is severely deficient.

We explicitly distinguish CRuSS from classical robustness summaries such as mean corruption accuracy, relative corruption error, or single-domain external F1. Those quantities answer useful but narrower questions. CRuSS asks whether a classifier jointly preserves predictive adequacy, cross-domain performance, probability calibration, global stress resilience, and class-local resilience. The empirical validation therefore focuses not only on absolute score values but also on rank reordering, component ablation, weighting sensitivity, and provenance-aware interpretation.

3. Materials and methods

3.1. Design criteria and pre-specified hypotheses

Before computing the final score, we defined six desiderata. *Boundedness*: every component and the final index must have an interpretable finite range. *Monotonicity*: improving any component while holding the others fixed must never lower the score. *Bottleneck sensitivity*: a severe failure in one axis should remain visible after aggregation. *Scale compatibility*: components should be dimensionless or normalized before aggregation. *Auditability*: the scalar must decompose back into clinically interpretable axes. *No outcome-fitted weights*: default weights must not be optimized to favor any architecture.

The validation was organized around four pre-specified hypotheses. H1: clean macro-F1 ranking will not fully reproduce CRuSS ranking because the stress components contribute non-redundant information. H2: directional source-to-target transfer will be asymmetric, motivating a directed rather than pooled generalization component. H3: global and class-local corruption resilience will identify different vulnerabilities. H4: the principal ranking, particularly the leading model, will remain stable under plausible weight perturbations and leave-one-component-out analysis. These hypotheses convert the metric paper from a descriptive score proposal into an empirical construct-validation study.

3.2. Datasets and provenance

Three four-class brain-MRI collections were used as source domains: 7K (7,023 images), 10K (10,417 images), and 40K (40,100 images). All were mapped to the canonical class order glioma, meningioma, no-tumor, and pituitary. The 7K collection was obtained from the public Kaggle release "Brain Tumor (MRI Scans)" (Rajarshi Mandal; `https://www.kaggle.com/datasets/rm1000/brain-tumor-mri-scans`). The 10K collection corresponds to

the Mendeley Data version-2 release by Siam and Khan [25]. The 40K collection was obtained from the public Kaggle release "Denoised MRI Brain Tumor Dataset 40K" (Lamyaa Fahim; https://www.kaggle.com/datasets/lamyaafahim/denoised-mri-brain-tumor-dataset-40k) and is associated with the denoised/augmented brain-MRI study of Katran et al. [27]. Because patient identifiers are not consistently available across these public releases, dataset labels are not interpreted as proof of independent patient cohorts.

The fixed splits used for training/validation/test were 4,916/1,053/1,054 for 7K; 7,291/1,563/1,563 for 10K; and 28,069/6,015/6,016 for 40K. Patient identifiers were not available consistently, so patient-level independence could not be asserted. An empirical duplicate audit over 57,540 represented images found 48 exact SHA256 cross-dataset pairs, all between 10K and 40K, and a large number of pHash ≤ 4 candidates. These pHash candidates were treated as screening signals, not automatically as ground-truth duplicates.

A fourth, operationally unseen evaluation source was drawn from the public PMRAM: Bangladeshi Brain Cancer MRI dataset [26] (Mendeley Data, version 1; DOI: 10.17632/m7w55sw88b.1). The public release contains 1,600 raw images and 6,000 images after augmentation; the project-specific evaluation manifest contained 1,505 images. After evaluation, this manifest was subjected to a deeper near-duplicate audit against the three development collections. Strong Tier A/B transformed-same-slice evidence was found for 787 of the 1,505 evaluated images; notably, all 373 pituitary images in the evaluated manifest were Tier A/B candidates. PMRAM was therefore excluded from claims of independent external generalization and retained only as a provenance sensitivity analysis. We use "unseen" here only in the operational sense that this fourth source was not used for model training or selection; the overlap audit prevents interpreting it as an independent patient-level external cohort.

Table 1
Dataset composition and fixed split sizes used in the main study.

Dataset	Total	Train	Validation	Test	Provenance status
7K	7,023	4,916	1,053	1,054	Public Kaggle release
10K	10,417	7,291	1,563	1,563	Mendeley Data v2 public release
40K	40,100	28,069	6,015	6,016	Public denoised/augmented Kaggle release
PMRAM evaluated manifest	1,505	–	–	1,505	Operationally unseen; audit-disqualified as independent

3.3. Model families and repeated runs

The benchmark includes three CNNs (ResNet50, DenseNet121, MobileNetV2), three supervised vision transformers (ViT-B/16, DeiT-Small, Swin-Tiny), a vision–language encoder (CLIP ViT-B/32), and a self-supervised foundation encoder (DINOv2-Base) [17, 18, 19, 20, 21, 22, 23, 24]. Each model was trained on each source dataset using seeds 42, 123, 777, 2026, and 3407, producing 120 final checkpoints. The purpose of architectural diversity is construct validation: a reliability metric should discriminate among qualitatively different inductive biases rather than reproduce one architecture family's clean ranking.

Table 2
Model-family diversity used to probe the reliability construct. The benchmark intentionally spans distinct representation and pretraining paradigms.

Model	Family	Representation / pretraining paradigm	Evaluation normalization
ResNet50	CNN	Residual convolutional hierarchy	ImageNet
DenseNet121	CNN	Dense feature reuse across convolutional blocks	ImageNet
MobileNetV2	CNN	Lightweight inverted residual bottlenecks	ImageNet
ViT-B/16	Transformer	Patch-token global self-attention	ImageNet
DeiT-Small	Transformer	Data-efficient supervised transformer	ImageNet
Swin-Tiny	Transformer	Hierarchical shifted-window attention	ImageNet
CLIP ViT-B/32	Vision–language	Contrastive image–text pretraining	CLIP native
DINOv2-Base	Self-supervised ViT	Large-scale self-distillation / SSL features	ImageNet-style

All inputs were converted to RGB and resized to 224×224 using bilinear interpolation for cross-domain inference. No target-specific normalization, threshold tuning, test-time augmentation, or validation-set selection was permitted.

Keeping the source-model preprocessing unchanged is necessary for the cross-domain experiment to measure deployment transfer rather than adaptation ability.

3.4. Cross-domain protocol

Every final checkpoint was evaluated without adaptation on the other two datasets, resulting in 240 source-to-target runs. There was no target retraining, fine-tuning, validation tuning, target-specific threshold selection, or test-time augmentation. Source-model preprocessing was preserved. Each run reports accuracy, balanced accuracy, macro/weighted F1, MCC, Cohen's kappa, macro AUROC, macro AUPRC, ECE, negative log-likelihood, and multiclass Brier score.

3.5. Controlled corruption benchmark

A fixed 7K seed-42 test set of 1,054 images was used for progressive stress testing. Three corruption families were applied: Gaussian noise, Gaussian blur, and brightness/contrast perturbation. Severity 0 denotes the unmodified image, and severities 1–5 increase perturbation strength. All eight seed-42 models were evaluated at all six severity levels, producing 144 model–corruption–severity conditions. This design follows the principle of progressive corruption evaluation rather than evaluating only one arbitrary perturbation magnitude [9].

Table 3
Deterministic corruption severity schedule used for the stress benchmark. Noise values are standard deviations in $[0, 1]$ pixel space.

Severity	Noise σ	Blur σ	Brightness / contrast
0	0.00	0.0	1.00 / 1.00
1	0.02	0.5	1.10 / 0.95
2	0.04	1.0	1.20 / 0.90
3	0.06	1.5	1.30 / 0.80
4	0.08	2.0	1.40 / 0.70
5	0.10	3.0	1.50 / 0.60

Gaussian-noise realizations were deterministic and image-specific, using a fixed benchmark seed of 20261005; the same standard-normal realization was scaled across severity levels so that changes across the curve reflect severity rather than a new random draw. For Gaussian blur, kernel size followed $2\lceil 3\sigma \rceil + 1$. Brightness was applied before contrast. Images were resized to 224×224 with bilinear interpolation. CNN, ViT, DeiT, Swin, and DINOv2 models used ImageNet normalization $(0.485, 0.456, 0.406)/(0.229, 0.224, 0.225)$; CLIP used its native normalization. No augmentation or test-time augmentation was applied during evaluation. These details are included because stress metrics are only reproducible when perturbation schedules are fully specified.

3.6. Class-specific shift benchmark

To expose failures that macro-averaging can hide, we repeated the same three corruption families while perturbing only the meningioma images in the fixed 7K test set (247 images); glioma, no-tumor, and pituitary images (807 total) were unchanged. Target-class precision, recall, F1, AUROC, and AUPRC were recorded alongside overall metrics. This benchmark contains 144 conditions and allows explicit measurement of local-class retention.

3.7. Definition of CRuSS

CRuSS is built from five components in $[0, 1]$. Let m denote a model family, d a source dataset, t a target dataset, s a seed, c a corruption family, and $k \in \{0, \dots, 5\}$ a severity.

Clean adequacy. Let $F^{\mathrm{ID}}_{m,d,s}$ be clean macro-F1. We define

$$A_m = \exp\left(\frac{1}{|\mathcal{D}||\mathcal{S}|} \sum_{d,s} \log F^{\mathrm{ID}}_{m,d,s} \right). \tag{1}$$

The geometric average reduces domination by any single dataset while preserving the native bounded scale.

Cross-domain retention. For each no-adaptation source-to-target run,

$$r^{\mathrm{OOD}}_{m,d\to t,s} = \min\left(1, \frac{F^{\mathrm{OOD}}_{m,d\to t,s}}{F^{\mathrm{ID}}_{m,d,s}}\right), \tag{2}$$

and

$$G_m = \exp\left(\frac{1}{N_{\mathrm{ext}}} \sum \log r^{\mathrm{OOD}}\right). \tag{3}$$

Clipping at one treats improvement on an easier target as full retention rather than a robustness bonus.

Calibration quality under shift. With $\mathrm{ECE}_{m,i} \in [0, 1]$ over the external evaluations,

$$Q_m = 1 - \frac{1}{N_{\mathrm{ext}}} \sum_i \mathrm{ECE}_{m,i}. \tag{4}$$

This component rewards probability reliability without introducing an arbitrary temperature or scaling constant.

Global corruption resilience. For each corruption c, macro-F1 retention is

$$r^{\mathrm{corr}}_{m,c}(k) = \min\left(1, \frac{F_{m,c}(k)}{F_{m,c}(0)}\right). \tag{5}$$

The normalized trapezoidal area over severity is

$$R_{m,c} = \frac{1}{5} \int_0^5 r^{\mathrm{corr}}_{m,c}(k)\, dk, \tag{6}$$

and R_m is the geometric mean across the three corruption families.

Class-local resilience. Replacing global macro-F1 with meningioma F1 under meningioma-only perturbation yields $L_{m,c}$ and

$$L_m = \left(\prod_c L_{m,c}\right)^{1/|C|}. \tag{7}$$

Final score. The primary score is

$$\boxed{\mathrm{CRuSS}_m = 100\,(A_m G_m Q_m R_m L_m)^{1/5}.} \tag{8}$$

The score is bounded in $[0, 100]$, monotonic in every component, equals 100 only when every component equals one, and is bottleneck-sensitive because the geometric mean cannot completely compensate a low component with a saturated one. Equal log-space weighting is the default; no weights are fit to outcomes.

3.8. Aggregation rationale and mathematical properties

Let $\mathbf{C}_m = (A_m, G_m, Q_m, R_m, L_m)$ denote the five-component reliability vector for model m. The equal-weight geometric mean can be written in log space as

$$\log(\mathrm{CRuSS}_m/100) = \frac{1}{5} \sum_{j=1}^{5} \log C_{m,j}. \tag{9}$$

This representation makes three properties explicit. First, the score is monotonic because $\partial \mathrm{CRuSS}/\partial C_j > 0$ for $C_j > 0$. Second, the relative influence of a component grows as that component becomes small: $\partial \log \mathrm{CRuSS}/\partial C_j = 1/(5C_j)$. Third, equal weights correspond to equal contributions in log space rather than equal additive percentage points.

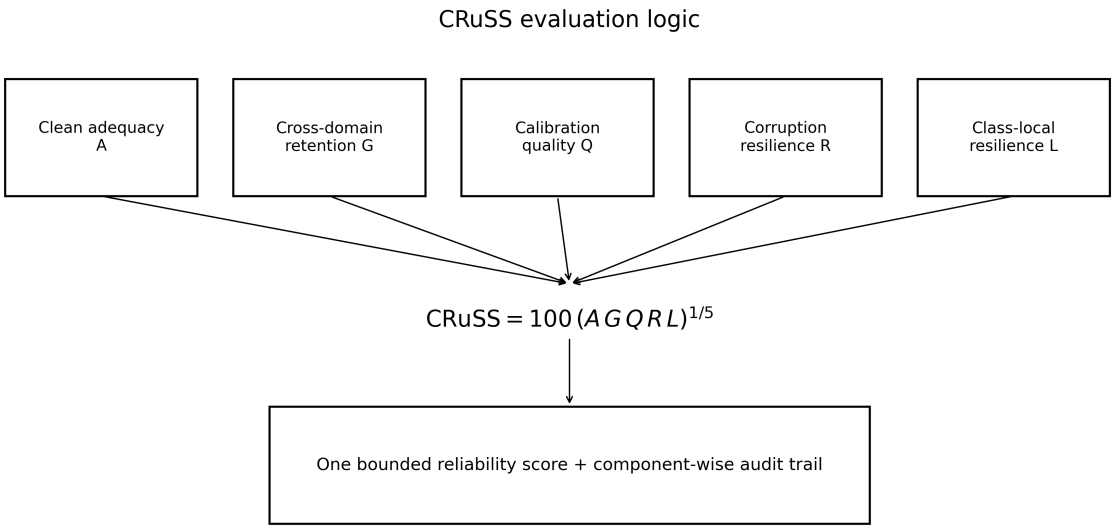


Figure 1: CRuSS evaluation logic. Five bounded axes are retained as an audit trail and aggregated geometrically into one reliability score.

Consequently, moving a weak component from 0.70 to 0.80 has more influence than moving an already saturated component from 0.98 to 0.99, which is desirable for a bottleneck-sensitive reliability index.

The clipping operation in the retention terms is intentional. When a target domain is easier than the source domain, a ratio above one is not interpreted as super-robustness; it is treated as full retention. Without clipping, an unusually easy target could compensate for degradation on another target. Likewise, the corruption terms integrate the complete severity trajectory using a normalized trapezoidal area rather than selecting a single endpoint. This reduces dependence on an arbitrary severity choice and rewards graceful degradation.

3.9. Why macro-F1 anchors adequacy and retention

The four-class problem is not perfectly balanced and the clinical concern is class-wise performance rather than only majority-class correctness. Macro-F1 therefore provides a direct class-balanced summary of precision and recall and is used as the anchor for clean adequacy and retention. Accuracy, balanced accuracy, MCC, AUROC, and AUPRC are still reported and used in convergent-validity analyses. We do not claim that macro-F1 is universally optimal; for a different task, the CRuSS framework can substitute another pre-declared task-valid base metric, consistent with the problem-aware philosophy of Metrics Reloaded [1].

3.10. Calibration component and its scope

The calibration component uses cross-domain ECE because probability reliability is most relevant when the input distribution differs from development data. ECE is simple and interpretable but binning-dependent, so we avoid presenting Q as a complete uncertainty characterization. NLL and multiclass Brier score are retained in the source-to-target result tables as secondary calibration-sensitive quantities. A future implementation could replace Q with a geometric combination of normalized ECE and Brier skill if a clinically validated reference distribution were available; this was deliberately not done here to avoid introducing additional scaling constants.

3.11. Leakage control and provenance eligibility

A reliability score cannot repair invalid test-set provenance. For that reason, provenance checking is conceptually upstream of CRuSS. Exact SHA256 matching was used to detect byte-identical files. Perceptual hashing was then used as a high-recall candidate screen for transformed duplicates, followed by stronger structural/pixel/edge similarity checks in the PMRAM audit. Candidate overlap was not automatically labeled as patient identity; rather, strong transformed-same-slice evidence was treated as sufficient to invalidate claims of independent external generalization

when class coverage would collapse after exclusion. This eligibility rule prevents an apparently high external score from being interpreted as evidence of deployment generalization when the benchmark itself is compromised.

3.12. Uncertainty and sensitivity analyses

CRuSS uncertainty was quantified with 5,000 environment-level bootstrap replicates. For each model, clean model–dataset–seed runs were resampled for A; source-to-target runs were jointly resampled for G and Q; and corruption families were resampled for R and L. This interval quantifies sensitivity to evaluated environments and seeds, not patient-level sampling uncertainty.

To test dependence on equal weighting, 10,000 weight vectors were drawn from a symmetric Dirichlet distribution. For a weight vector $\mathbf{w}$, the generalized score was $100\exp(\sum_j w_j \log C_j)$, where $C_j \in \{A, G, Q, R, L\}$. Ranking agreement with the equal-weight score was measured by Kendall's τ. We additionally performed leave-one-component-out ranking analysis.

3.13. Construct-validation strategy and decision rules

CRuSS is intended to represent a latent practical construct—reliability under deployment-relevant shift—rather than to reproduce any single observed metric. We therefore evaluated the score using four complementary forms of construct evidence. *Convergent evidence* requires CRuSS to increase when external discrimination and calibration improve. *Discriminant evidence* requires the score not to collapse into a surrogate for clean macro-F1 or accuracy. *Structural evidence* requires the five axes to remain sufficiently non-redundant that each contributes distinct information. Finally, *consequence evidence* asks whether the score changes model selection in cases where clean metrics are nearly saturated but stress behavior differs. These criteria were specified before interpretation of the final ranking and are reported jointly rather than selecting only favorable analyses.

We also distinguish score construction from score use. CRuSS is a comparative index, not a clinically validated safety threshold. A value of 95 is not interpreted as "95% safe," and no universal accept/reject cut-off is proposed. For a deployment study, the five components and their uncertainty intervals should be examined alongside the aggregate score, with domain-specific minimum requirements defined prospectively. This distinction prevents a convenient ranking tool from being over-interpreted as an absolute guarantee of clinical readiness.

3.14. Pareto and dominance analysis

A scalar score is most useful when its ranking can be reconciled with the underlying multi-objective problem. We therefore performed Pareto-dominance analysis on the five-dimensional vector (A, G, Q, R, L). Model i was said to dominate model j only if it was no worse on every component and strictly better on at least one. This analysis is independent of the geometric aggregation rule and provides a weight-free check on whether a highly ranked model is broadly strong or is merely favored by compensation among components.

3.15. Directional asymmetry and degradation-rate analysis

External shift is not assumed to be symmetric. For each unordered dataset pair, we computed the absolute difference between mean macro-F1 in the two transfer directions. We summarize this as directional asymmetry because training on dataset a and testing on dataset b need not be equivalent to the reverse experiment. For progressive corruptions, we additionally fit a simple least-squares slope to normalized macro-F1 retention across severities 0–5. The slope is descriptive rather than causal, but it complements area-under-the-retention-curve analysis by exposing how rapidly performance deteriorates as stress increases.

3.16. Statistical reporting

We report mean and standard deviation across seeds where relevant, environment-bootstrap 95% intervals for CRuSS, Spearman rank correlation between CRuSS and conventional aggregate metrics, and rank inversions. Given only eight model families, correlation p-values are interpreted descriptively rather than as definitive evidence. The reporting philosophy follows transparent prediction-model evaluation principles [28, 29].

3.17. Reproducibility and analysis implementation

All metric calculations were performed from saved prediction/result tables rather than manually transcribed manuscript values. The clean table contains 120 model–dataset–seed rows and the directed cross-domain table contains 240 evaluations. Corruption and class-specific stress tables each contain 144 model–corruption–severity conditions. The manuscript package stores these derived tables together with the score summary, weight-sensitivity results, and

provenance summary. The calculation sequence is deterministic given these files: compute the five components, aggregate using Eq. 8, bootstrap evaluated environments, then perform rank and sensitivity analyses. Missing NLL/Brier entries in some clean runs are not used in CRuSS and therefore cannot change the primary score.

To reduce researcher degrees of freedom, no architecture-specific thresholds, component-specific learned coefficients, or target-domain tuning were introduced after observing the final ranking. The same component definitions are applied to all eight model families. This is important because a metric intended for benchmarking should not depend on hidden model-specific calibration choices.

4. Results

4.1. Clean performance was uniformly high

Across the 120 clean runs, mean macro-F1 ranged from 97.36% (DeiT-Small) to 99.27% (DINOv2-Base). MobileNetV2 ranked second by clean macro-F1 (98.90%), followed by ResNet50 (98.82%) and DenseNet121 (98.63%). The narrow clean range illustrates why a single near-saturated benchmark metric provides limited information about deployment stress behavior.

Table 4
CRuSS components and final score. Values are model-level aggregates in the unit interval except CRuSS, which is shown on a 0–100 scale.

Model	A	G	Q	R	L	CRuSS
DINOv2-Base	0.9927	0.9952	0.9935	0.9524	0.9842	**98.35**
ViT-B/16	0.9838	0.9894	0.9889	0.9106	0.9564	96.54
CLIP ViT-B/32	0.9831	0.9847	0.9859	0.8966	0.9418	95.78
ResNet50	0.9881	0.9872	0.9914	0.8684	0.9316	95.21
DeiT-Small	0.9735	0.9725	0.9817	0.8895	0.9325	94.93
Swin-Tiny	0.9837	0.9829	0.9880	0.8125	0.9676	94.43
DenseNet121	0.9863	0.9790	0.9877	0.8471	0.7962	91.55
MobileNetV2	0.9889	0.9837	0.9896	0.7838	0.7956	90.30

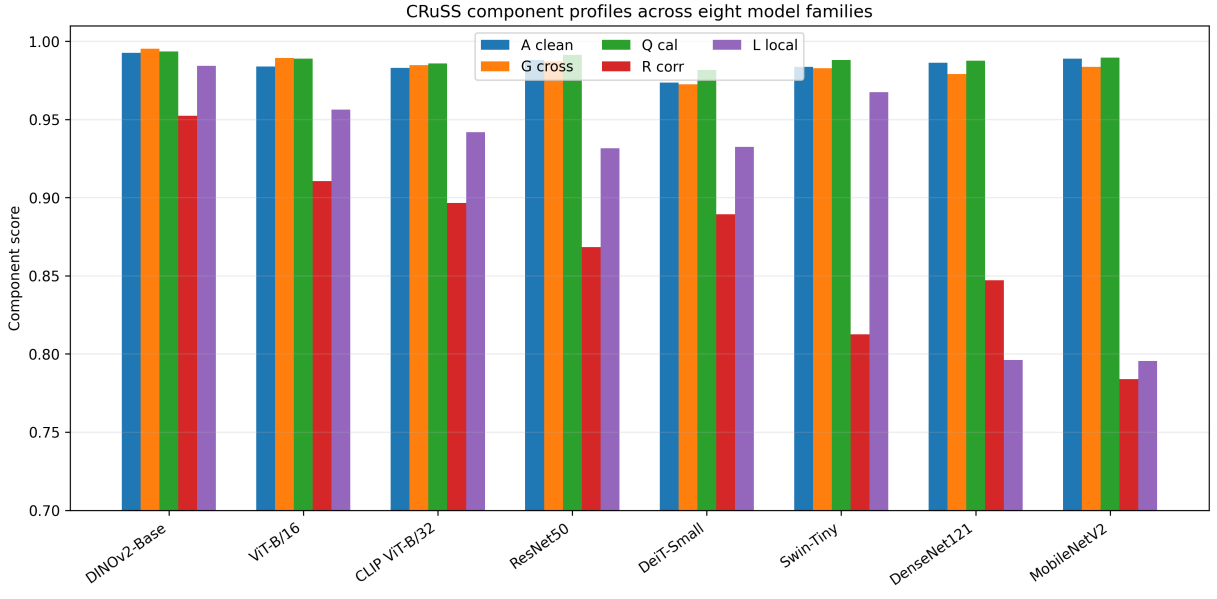


Figure 2: Component profiles show that the largest between-model separation arises from corruption resilience and class-local resilience, not clean adequacy or calibration.

4.2. CRuSS produced substantial rank inversions

The clean ranking and reliability ranking differed markedly. MobileNetV2 moved from rank 2 by mean clean macro-F1 to rank 8 by CRuSS because its global corruption and class-local components were the lowest among the eight models. Conversely, CLIP ViT-B/32 moved from rank 7 by clean macro-F1 to rank 3 by CRuSS. ViT-B/16 moved from rank 5 to rank 2. Across the eight models, Spearman correlation between clean macro-F1 and CRuSS was $\rho = 0.048$ ($p = 0.911$). This result should not be overinterpreted as a population-level null association; rather, it demonstrates that the selected model set contains strong rank disagreement.

Table 5
Rank inversion between clean macro-F1 and CRuSS.

Model	Clean macro-F1	Clean rank	CRuSS	CRuSS rank
DINOv2-Base	0.9927	1	98.35	1
MobileNetV2	0.9890	2	90.30	8
ResNet50	0.9882	3	95.21	4
DenseNet121	0.9863	4	91.55	7
ViT-B/16	0.9839	5	96.54	2
Swin-Tiny	0.9838	6	94.43	6
CLIP ViT-B/32	0.9831	7	95.78	3
DeiT-Small	0.9736	8	94.93	5

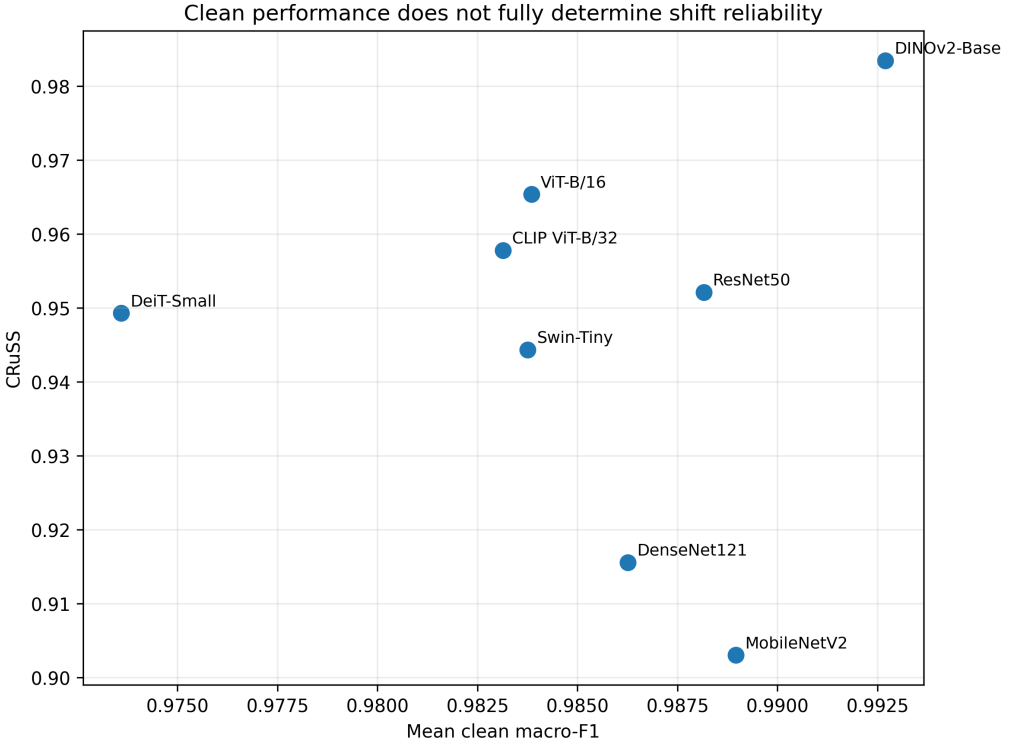


Figure 3: Mean clean macro-F1 versus CRuSS. Near-saturated clean performance does not uniquely determine stress reliability.

4.3. Cross-domain generalization was directional

External macro-F1 depended strongly on the source–target direction. Averaged over models and seeds, 40K→7K achieved 0.9960 and 10K→7K achieved 0.9935, whereas 7K→40K and 10K→40K were lower at 0.9353 and 0.9418,

respectively. This asymmetry supports using retention over all directed source–target pairs rather than reporting one external dataset as universally representative.

Table 6
Directed cross-dataset macro-F1, averaged across eight models and five seeds.

Source → target	Mean macro-F1	SD
7K → 10K	0.9814	0.0053
7K → 40K	0.9353	0.0261
10K → 7K	0.9935	0.0038
10K → 40K	0.9418	0.0241
40K → 7K	0.9960	0.0026
40K → 10K	0.9881	0.0025

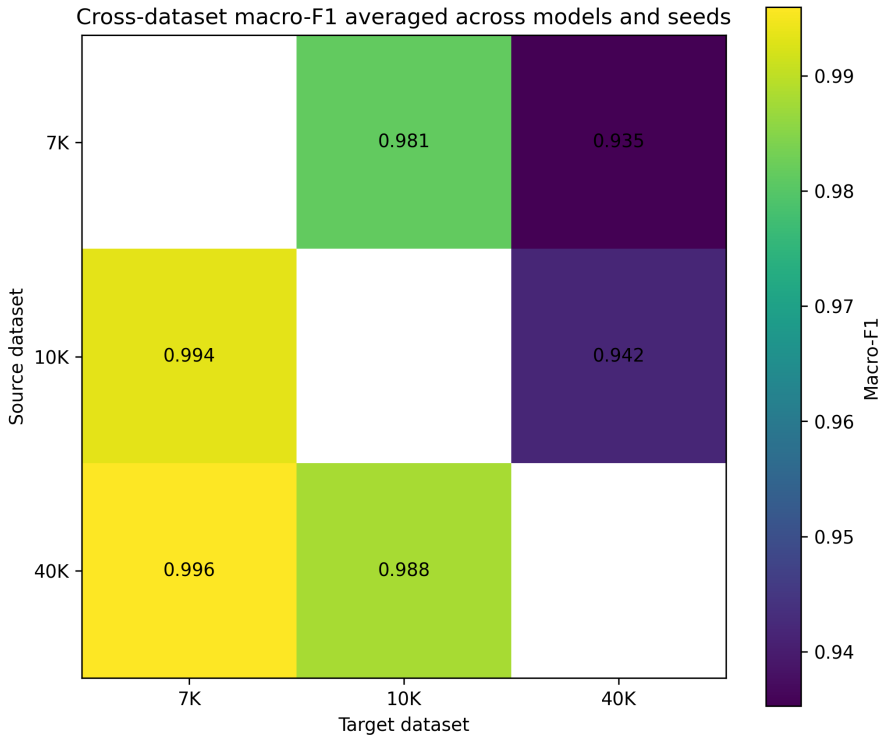


Figure 4: Directional cross-dataset macro-F1 averaged across models and seeds. The matrix is intentionally directed; source and target exchange is not assumed symmetric.

4.4. Progressive corruption separated models that looked similar when clean

At severity 5, mean macro-F1 across the three corruption families ranged from 0.610 for MobileNetV2 to 0.850 for DINOv2-Base. The full retention curves show nonuniform degradation: brightness/contrast was often mild, whereas blur and Gaussian noise caused substantially larger losses for several models. Because R integrates the entire severity trajectory, it is less sensitive to an arbitrary single severity than endpoint reporting.

4.5. Class-local stress exposed additional vulnerability

When only meningioma images were perturbed, severity-5 meningioma F1 averaged 0.940 for DINOv2-Base but only 0.613 for MobileNetV2. DenseNet121 also showed a low integrated local component ($L = 0.7962$) despite strong clean adequacy. This finding motivates keeping local-class resilience separate from global robustness: a model can maintain acceptable aggregate behavior while one diagnosis degrades disproportionately.

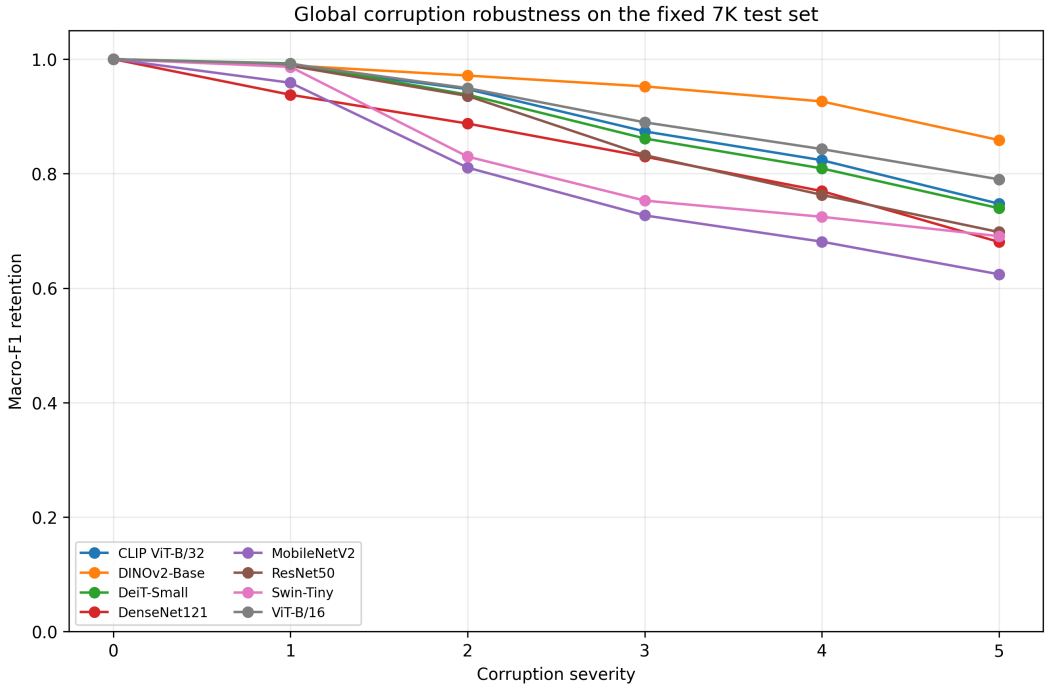


Figure 5: Global macro-F1 retention under progressive corruption, averaged over Gaussian noise, Gaussian blur, and brightness/contrast perturbation.

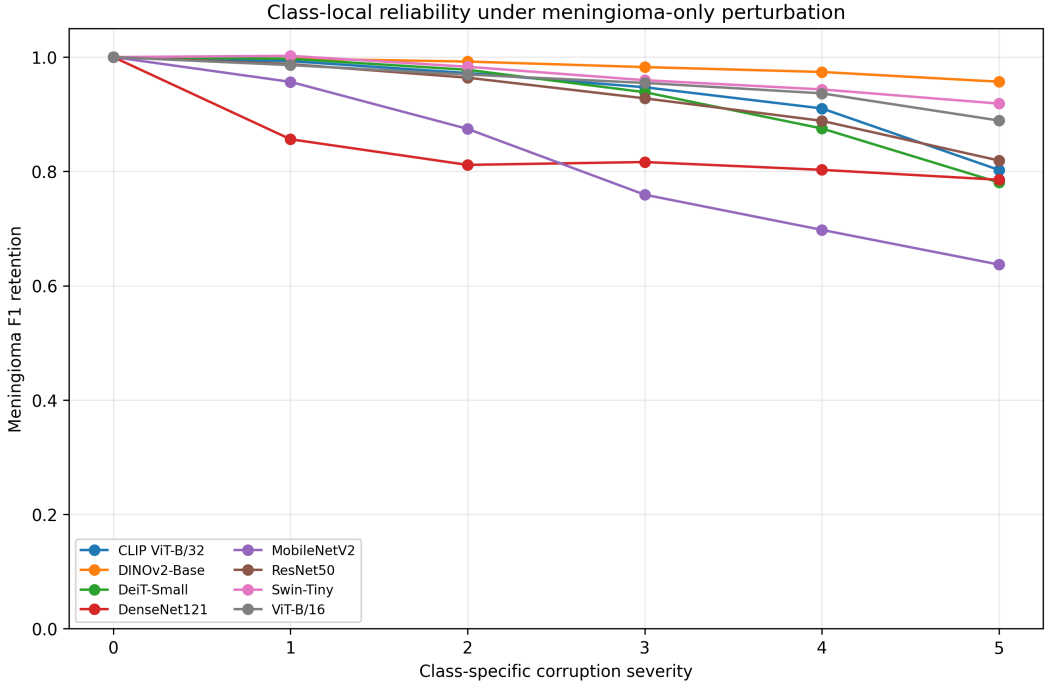


Figure 6: Meningioma F1 retention when only meningioma images are perturbed. Other classes remain unchanged, isolating class-local fragility.

4.6. Bootstrap uncertainty and weight sensitivity

Environment-bootstrap intervals were widest for the least stress-robust models. DINOv2-Base achieved CRuSS 98.35 with a 95% interval of 97.05–99.39; MobileNetV2 achieved 90.30 with an interval of 83.97–96.78. Over 10,000 symmetric Dirichlet weight draws, the mean Kendall rank agreement with the equal-weight score was 0.871; the median was 0.929 and the 2.5th percentile was 0.643. DINOv2-Base ranked first in all 10,000 draws, indicating that its top position was not an artifact of equal weights.

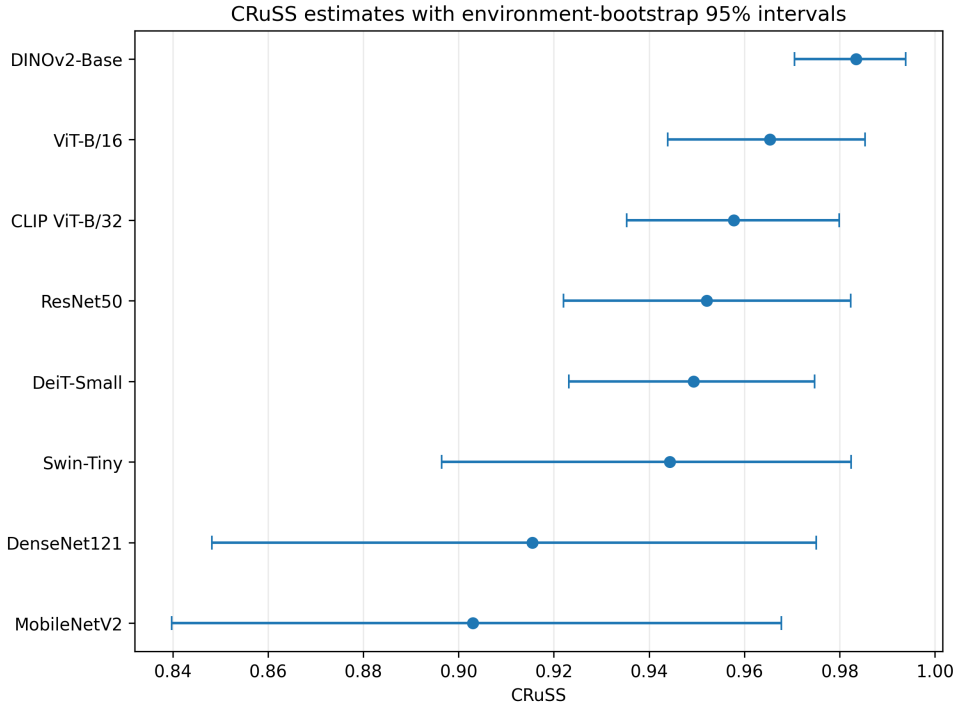


Figure 7: CRuSS point estimates and 95% environment-bootstrap intervals. These intervals quantify variation over evaluated environments/seeds, not patient-level uncertainty.

4.7. Provenance audit changed the interpretation of Dataset 4

The PMRAM experiment initially appeared suitable as an additional hidden external dataset. Exact SHA256 comparison alone found no byte-identical overlap, but a deeper audit using pHash screening and normalized structural/pixel/edge similarity found strong Tier A/B evidence for 787 of 1,505 images. The class pattern was particularly concerning: 373 of 373 pituitary images and approximately half of glioma/no-tumor images were Tier A/B candidates. Because filtering those candidates would eliminate the pituitary class, PMRAM could not remain a valid four-class independent test set. We therefore report the audit outcome but exclude PMRAM performance from CRuSS and from independent-generalization claims.

4.8. Conventional clean metrics were near-saturated and mutually redundant

Table 7 expands the clean-test summary beyond macro-F1. All eight models achieved mean AUROC above 0.998 and mean AUPRC above 0.994. Such saturation is useful evidence of task learnability but makes architecture selection difficult: the absolute spread in mean macro-F1 was only about 1.91 percentage points from the weakest to strongest model. The corresponding spread in CRuSS was 8.05 points on the 0–100 scale, indicating that stress evaluation created substantially greater separation.

4.9. Clean performance differed modestly by dataset

Although the pooled clean metrics were saturated, dataset-wise macro-F1 was not identical (Table 8). The 40K split was generally easiest for the trained models, while 7K produced lower values for several transformer families. Reporting

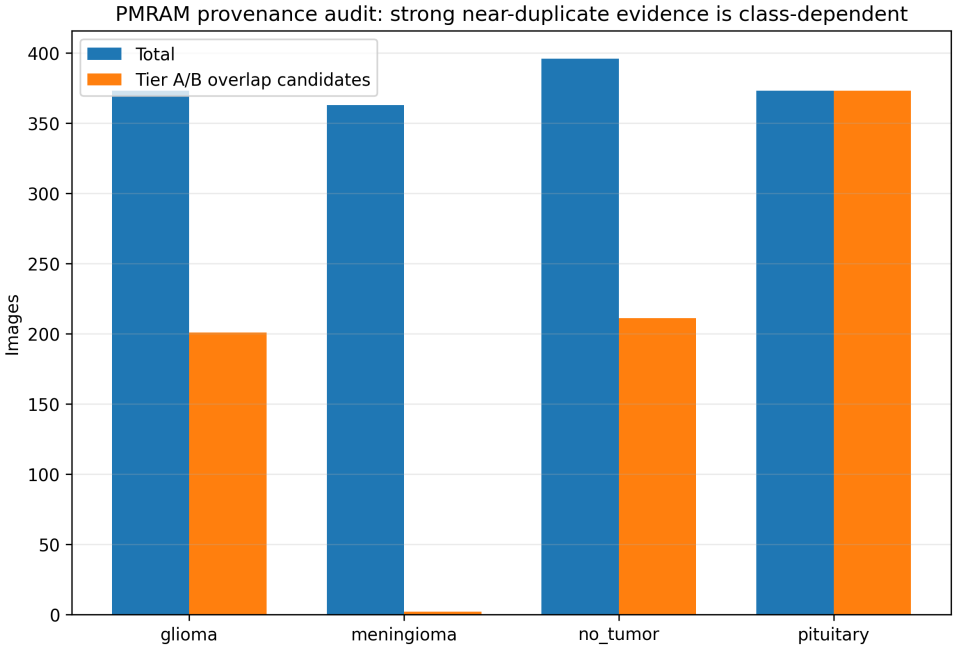


Figure 8: PMRAM overlap audit. Tier A/B candidates indicate strong transformed-same-slice evidence, not proven patient identity. The result was sufficient to disqualify PMRAM as an independent four-class external benchmark.

Table 7
Clean-test performance aggregated over three datasets and five seeds. Values are mean ± SD.

Model	Accuracy	Macro-F1	MCC	ECE
DINOv2-Base	0.9929±0.0045	0.9927±0.0046	0.9905±0.0060	0.0048±0.0032
MobileNetV2	0.9893±0.0058	0.9890±0.0058	0.9857±0.0077	0.0060±0.0031
ResNet50	0.9885±0.0075	0.9882±0.0076	0.9846±0.0099	0.0064±0.0040
DenseNet121	0.9867±0.0060	0.9863±0.0060	0.9822±0.0080	0.0070±0.0034
ViT-B/16	0.9843±0.0099	0.9839±0.0102	0.9791±0.0132	0.0082±0.0047
Swin-Tiny	0.9842±0.0092	0.9838±0.0094	0.9790±0.0123	0.0075±0.0050
CLIP ViT-B/32	0.9836±0.0083	0.9831±0.0084	0.9782±0.0111	0.0082±0.0042
DeiT-Small	0.9745±0.0114	0.9736±0.0115	0.9660±0.0151	0.0089±0.0052

this decomposition matters because the denominator of a retention score should reflect the actual source-domain baseline rather than one pooled clean value.

4.10. Model-specific cross-domain behavior confirmed directional heterogeneity

The aggregate direction table in Table 6 can hide architecture-specific differences. Figure 9 therefore resolves all six directed transfers for every model. The 40K source generally transferred well to 7K and 10K, whereas transfer into 40K was systematically harder. This pattern is consistent with the fact that target difficulty and dataset construction differ, and it demonstrates why one external dataset cannot be treated as a universal robustness oracle.

The full table shows two useful facts hidden by the overall mean. First, all models transferred strongly into 7K, suggesting that 7K is not the most stringent external target in this benchmark. Second, the spread between architectures widened sharply for 7K→40K and 10K→40K. For example, DINOv2-Base maintained macro-F1 near 0.98 in both directions, whereas DeiT-Small fell below 0.90. CRuSS uses retention relative to the source-domain baseline so that these differences are interpreted as preserved capability rather than absolute target difficulty alone.

Table 8
Mean clean macro-F1 by training/evaluation dataset across five seeds.

Model	7K	10K	40K
DINOv2-Base	0.9886	0.9917	0.9978
MobileNetV2	0.9834	0.9878	0.9957
ResNet50	0.9816	0.9854	0.9974
DenseNet121	0.9806	0.9845	0.9937
ViT-B/16	0.9726	0.9846	0.9944
Swin-Tiny	0.9722	0.9851	0.9940
CLIP ViT-B/32	0.9749	0.9831	0.9914
DeiT-Small	0.9606	0.9732	0.9871

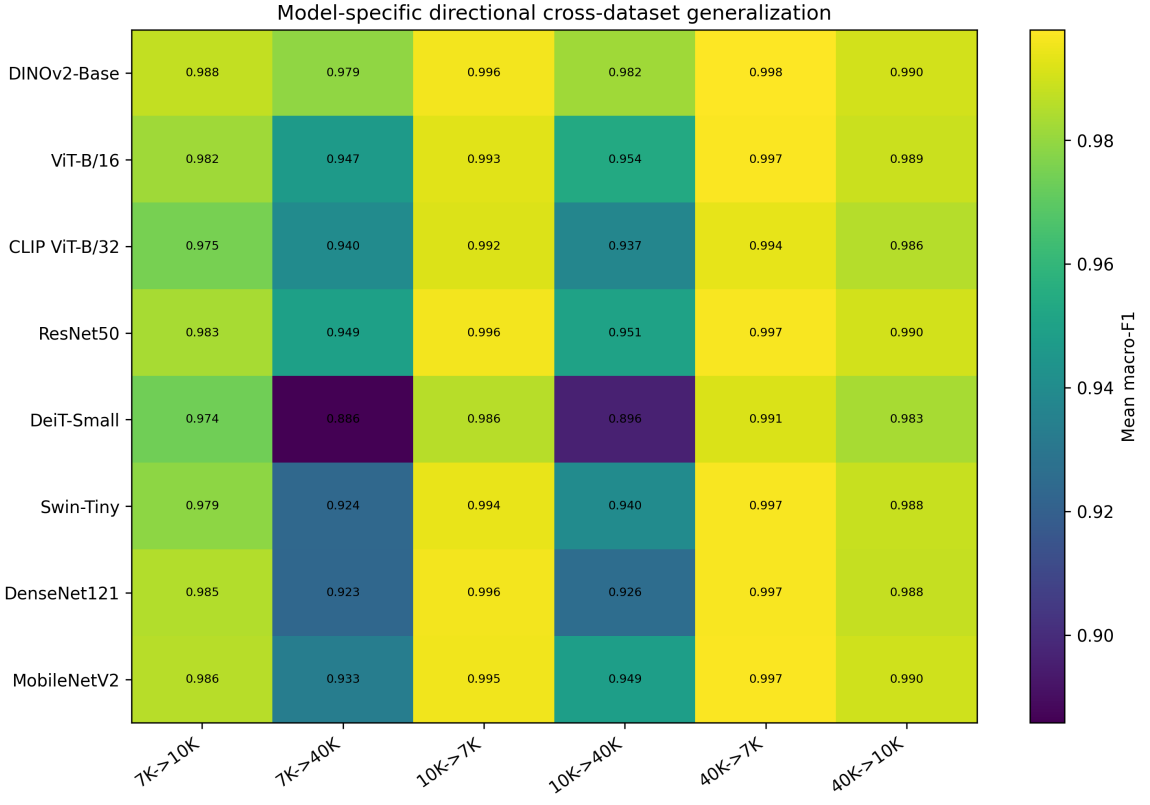


Figure 9: Model-specific directional cross-dataset macro-F1. Each cell averages five seeds; arrows indicate the training source and evaluation target.

Table 9
Mean macro-F1 for every model and directed source-to-target transfer, averaged over five seeds.

Model	7K→10K	7K→40K	10K→7K	10K→40K	40K→7K	40K→10K
DINOv2-Base	0.9876	0.9794	0.9960	0.9820	0.9979	0.9900
ResNet50	0.9833	0.9493	0.9958	0.9505	0.9968	0.9904
ViT-B/16	0.9820	0.9471	0.9930	0.9539	0.9974	0.9890
MobileNetV2	0.9859	0.9334	0.9954	0.9487	0.9970	0.9896
CLIP ViT-B/32	0.9749	0.9398	0.9918	0.9374	0.9938	0.9856
Swin-Tiny	0.9788	0.9239	0.9942	0.9400	0.9966	0.9885
DenseNet121	0.9850	0.9234	0.9960	0.9258	0.9968	0.9882
DeiT-Small	0.9738	0.8859	0.9859	0.8961	0.9914	0.9833

4.11. Calibration degraded under cross-domain deployment even when discrimination remained high

Across all models, mean ECE increased from clean to cross-domain evaluation. DINOv2-Base showed the smallest absolute increase (0.0017), while DeiT-Small showed the largest (0.0094). Figure 10 illustrates that calibration drift is not equivalent to clean accuracy ranking. This observation is aligned with prior medical-imaging evidence that acquisition shift can change clinically relevant operating behavior even when discrimination remains strong [30].

Table 10
Mean calibration error before and after no-adaptation cross-domain evaluation.

Model	Clean ECE	Cross-domain ECE	ΔECE
DINOv2-Base	0.0048	0.0065	0.0017
ResNet50	0.0064	0.0086	0.0021
ViT-B/16	0.0082	0.0111	0.0029
MobileNetV2	0.0060	0.0104	0.0044
Swin-Tiny	0.0075	0.0120	0.0045
DenseNet121	0.0070	0.0123	0.0053
CLIP ViT-B/32	0.0082	0.0141	0.0059
DeiT-Small	0.0089	0.0183	0.0094

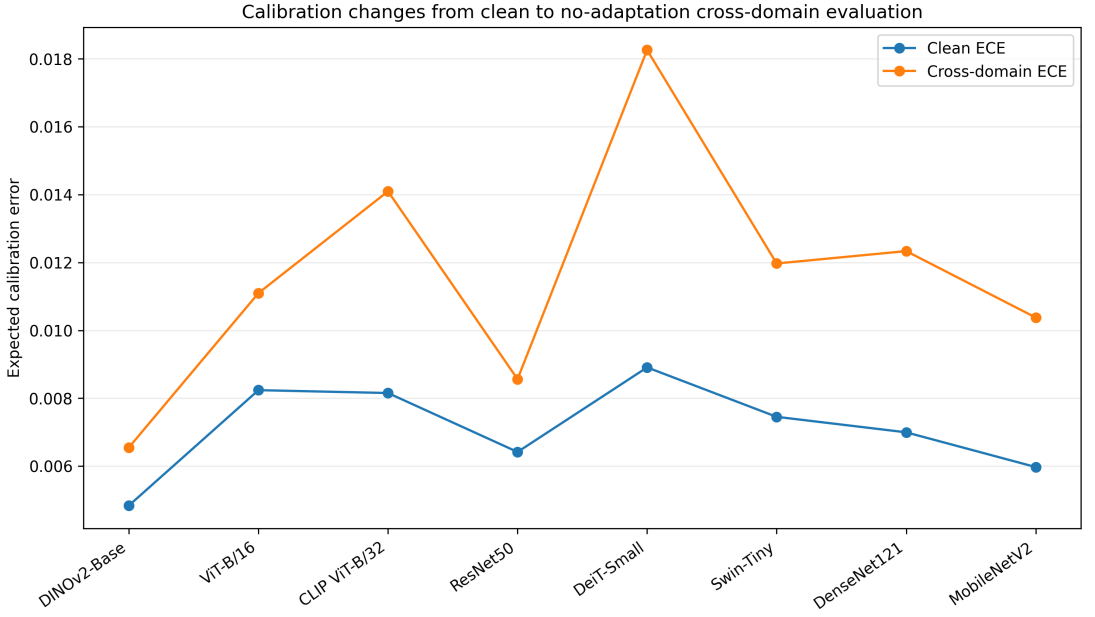


Figure 10: Clean and cross-domain ECE for each model family. A common rise in ECE supports treating calibration under shift as a separate reliability axis.

4.12. Corruption families contributed non-equivalent evidence

Brightness/contrast perturbation was comparatively mild for most models, with integrated retention close to one. Gaussian blur and Gaussian noise were substantially more discriminative. MobileNetV2 had a blur retention AUC of 0.631 and noise retention AUC of 0.765, while DINOv2-Base reached 0.971 and 0.891, respectively. ViT-B/16 was particularly strong under Gaussian noise (0.940). Figure 11 shows that R is not driven by a single corruption type; different architectures fail under different image degradations.

4.13. Global and class-local robustness were not interchangeable

The local stress benchmark produced failures that were not obvious from global corruption summaries. DenseNet121, for example, maintained respectable global resilience but its meningioma-only Gaussian-noise retention AUC fell to

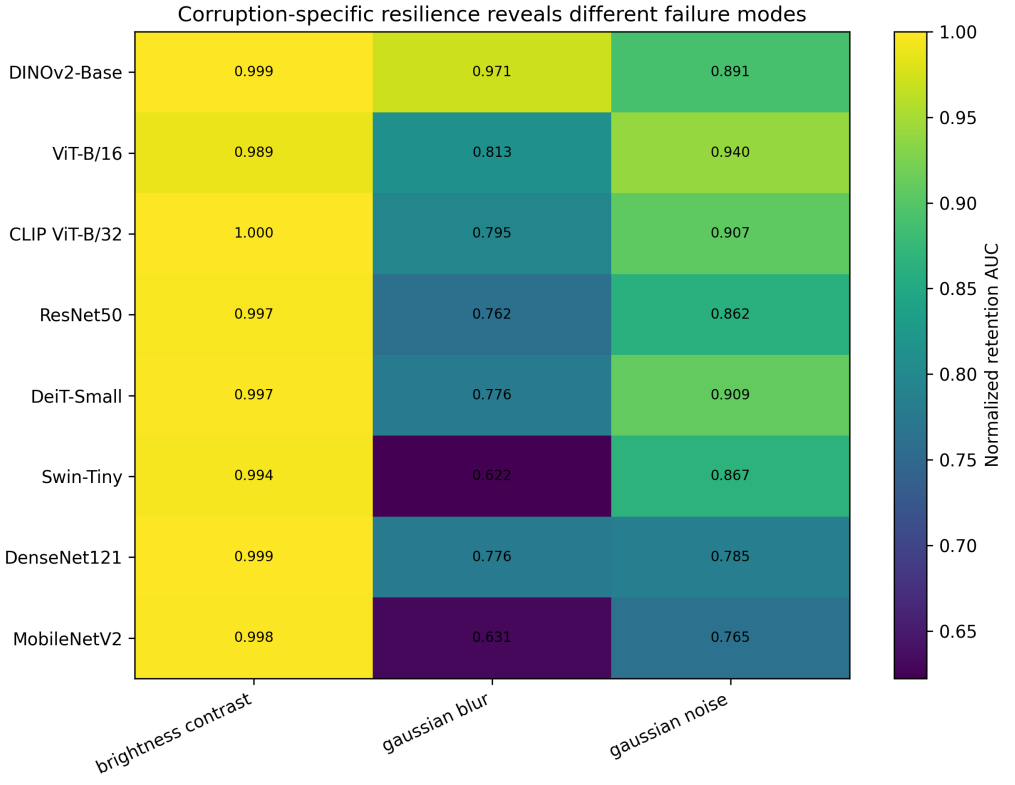


Figure 11: Normalized retention AUC for each corruption family. Values near one indicate graceful degradation across the complete severity trajectory.

0.505. MobileNetV2 showed a similar local weakness (0.553) despite near-saturated clean performance. In contrast, DINOv2-Base maintained class-local AUCs of 1.000, 0.996, and 0.957 for brightness/contrast, blur, and noise. This supports retaining L as an independent axis rather than assuming that a global robustness curve guarantees class-level safety.

Table 11
Class-local retention AUC under meningioma-only perturbation.

Model	Brightness/contrast	Gaussian blur	Gaussian noise
DINOv2-Base	1.000	0.996	0.957
Swin-Tiny	1.000	0.969	0.935
CLIP ViT-B/32	1.000	0.899	0.929
ViT-B/16	0.989	0.994	0.890
ResNet50	0.997	0.991	0.818
DeiT-Small	1.000	0.945	0.858
MobileNetV2	0.999	0.911	0.553
DenseNet121	1.000	0.999	0.505

4.14. Rank reordering is the primary evidence that CRuSS adds information

Figure 12 makes the clean-to-reliability rank changes explicit. MobileNetV2 drops six positions, CLIP rises four, and ViT-B/16 rises three. This is more informative than a correlation coefficient alone because the practical use case is model selection. The result supports H1: CRuSS is not merely a monotonic transformation of clean macro-F1 for the evaluated model set.

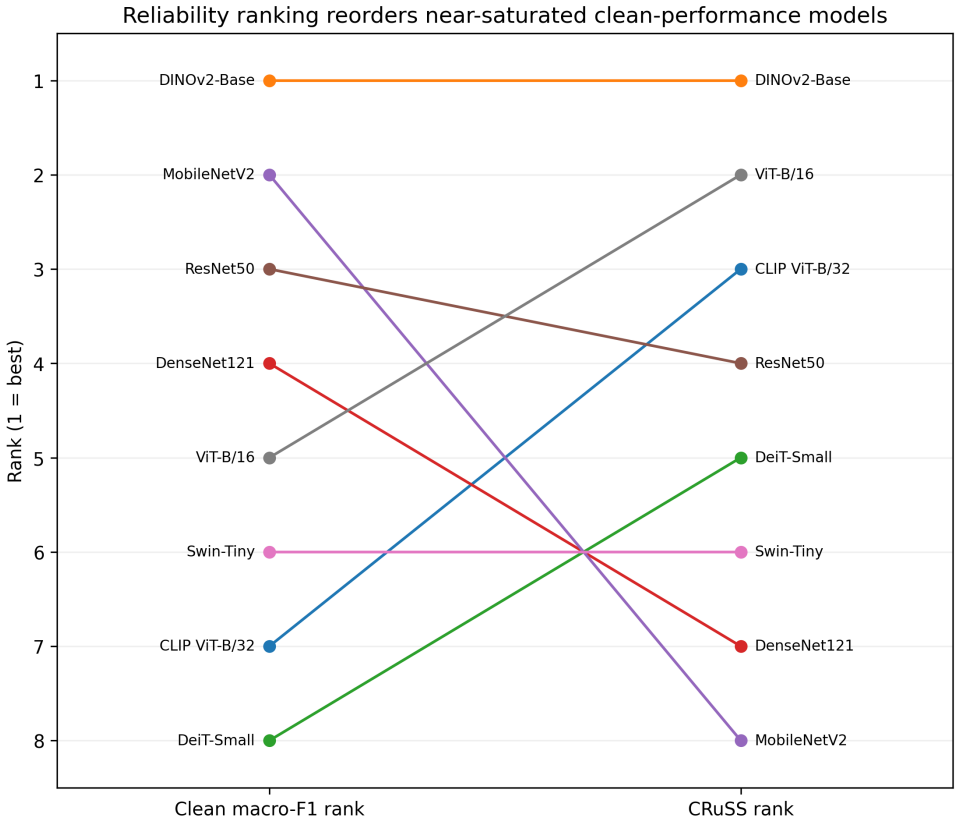


Figure 12: Slopegraph comparing model rank by clean macro-F1 and by CRuSS. Large crossings indicate non-redundant reliability information.

4.15. Association analysis supports convergent but non-redundant validity

CRuSS was essentially uncorrelated with clean accuracy, macro-F1, MCC, or AUROC in this eight-model sample, whereas its association with mean external F1 was positive ($\rho = 0.595$) and with mean external AUROC was $\rho = 0.643$. The small number of model families makes these estimates descriptive rather than inferential. The pattern is nevertheless useful for construct interpretation: the score is intentionally not a clean-performance surrogate, yet it moves in the expected direction with external performance.

Table 12
Spearman association between CRuSS and conventional aggregates across the eight model families. p-values are descriptive because $n = 8$.

Comparator	Spearman ρ	p
Clean accuracy	0.048	0.911
Clean macro-F1	0.048	0.911
Clean MCC	0.048	0.911
Clean macro-AUROC	0.000	1.000
Clean macro-AUPRC	-0.143	0.736
Clean ECE	0.071	0.867
Cross-domain macro-F1	0.595	0.120
Cross-domain macro-AUROC	0.643	0.086
Cross-domain ECE	-0.286	0.493

4.16. Ablation identified corruption resilience as the largest ranking driver

Leave-one-component-out analysis preserved strong agreement with the full ranking when A, G, or L was removed ($\tau = 0.929$), and removing Q left the ranking unchanged ($\tau = 1.000$). Removing global corruption resilience R reduced agreement to $\tau = 0.571$. This does not mean calibration is unimportant; rather, ECE values were uniformly small in this benchmark and therefore had limited ranking leverage. The ablation clarifies which component creates discrimination in the present data and prevents the scalar from being treated as a black box.

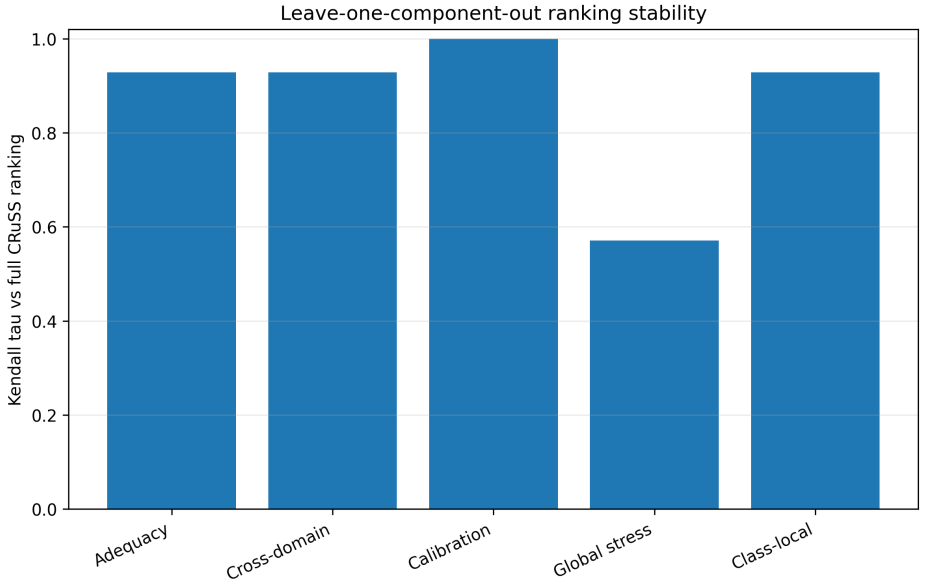


Figure 13: Kendall rank agreement between full CRuSS and leave-one-component-out variants. Lower agreement indicates that the omitted component contributes more to model ordering in this benchmark.

4.17. The main ranking did not depend on choosing the geometric operator

Because composite indices can be criticized for arbitrary aggregation, we compared the pre-specified geometric mean with three alternatives using the same five components: arithmetic mean, harmonic mean, and a strict minimum (weakest-link) operator. Arithmetic and harmonic aggregation reproduced the geometric ranking exactly for all eight models ($\tau = 1.000$). The minimum operator changed only the ordering of DeiT-Small and ResNet50, retaining high agreement ($\tau = 0.929$) and leaving DINOv2-Base first and MobileNetV2 last. Thus the headline model ordering is driven primarily by the observed component profiles rather than by a uniquely favorable mathematical operator.

Table 13
Sensitivity of model ranking to the aggregation operator. Rank 1 is best.

Model	Geometric	Arithmetic	Harmonic	Minimum
DINOv2-Base	1	1	1	1
ViT-B/16	2	2	2	2
CLIP ViT-B/32	3	3	3	3
ResNet50	4	4	4	5
DeiT-Small	5	5	5	4
Swin-Tiny	6	6	6	6
DenseNet121	7	7	7	7
MobileNetV2	8	8	8	8

The geometric mean remains our default because it provides smooth bottleneck sensitivity while preserving information from every axis. The arithmetic mean is more compensatory, the harmonic mean is more strongly influenced

by small values, and the minimum operator discards all information except the weakest component. The empirical operator sensitivity analysis shows that this theoretical preference is not being used to manufacture the reported ranking.

4.18. Random weighting rarely changed the main conclusion

The distribution of Kendall agreement over 10,000 symmetric Dirichlet weight draws is shown in Fig. 14. Most random weight choices preserved a ranking close to the equal-weight default, and DINOv2-Base remained first in every draw. This analysis addresses the common criticism that composite indices can be tuned to produce a preferred ranking. The result does not establish universal weight invariance, but it demonstrates that the headline conclusion is not fragile to moderate reweighting of the five measured axes.

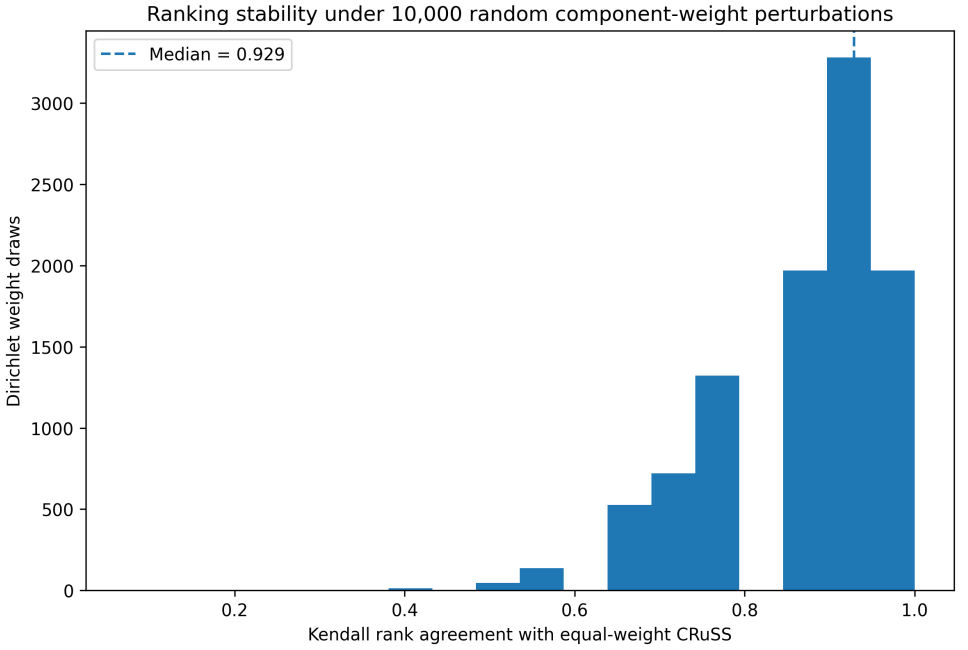


Figure 14: Distribution of Kendall rank agreement under 10,000 random Dirichlet component-weight draws.

4.19. Hypothesis-level summary

The four pre-specified hypotheses were supported descriptively. H1 was supported by substantial clean-to-CRuSS rank reordering and near-zero rank correlation with clean macro-F1. H2 was supported by pronounced source–target asymmetry, especially for transfer into 40K. H3 was supported by architecture-specific divergence between global and class-local corruption behavior. H4 was supported by high median Kendall agreement under random weighting and by stability of the leading model under all sampled weight vectors. These results do not establish universal validity, but they provide a stronger validation basis than simply presenting a new score on one dataset.

4.20. The five reliability axes were related but not redundant

Figure 15 reports pairwise Spearman correlations among the five CRuSS components. Clean adequacy was strongly associated with calibration quality ($\rho = 0.905$), which is expected because confident correct predictions tend to reduce calibration error in these high-performing models. In contrast, clean adequacy had essentially no monotonic relationship with global corruption resilience ($\rho = -0.024$) or class-local resilience ($\rho = -0.095$). Global and class-local resilience were moderately correlated ($\rho = 0.643$), indicating shared sensitivity to image degradation without equivalence. These patterns support the structural rationale for retaining separate stress components: the information provided by R and L is not recoverable from clean performance alone.

Table 14
Spearman correlation matrix among the five CRuSS components across the eight evaluated model families.

	A	G	Q	R	L
A	1.000	0.571	0.905	-0.024	-0.095
G	0.571	1.000	0.762	0.619	0.476
Q	0.905	0.762	1.000	0.119	0.167
R	-0.024	0.619	0.119	1.000	0.643
L	-0.095	0.476	0.167	0.643	1.000

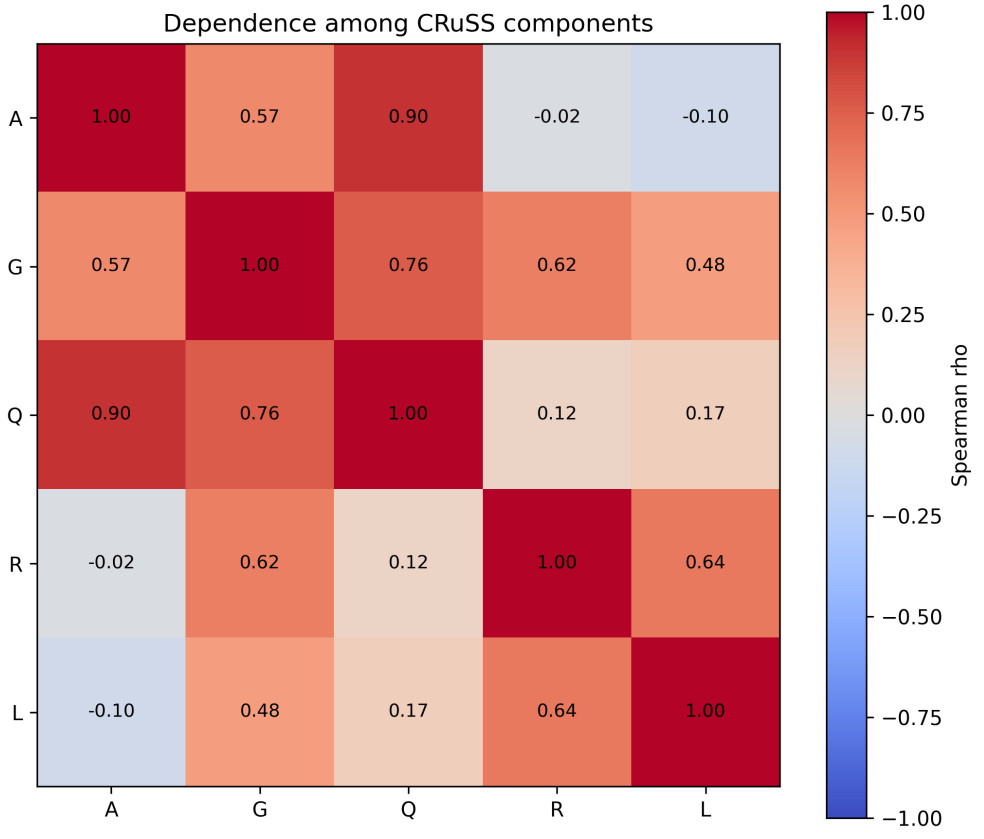


Figure 15: Dependence among CRuSS components. The stress-resilience axes contain information that is weakly related to clean adequacy.

4.21. Cross-dataset transfer showed pronounced directional asymmetry

Averaging absolute forward-versus-reverse macro-F1 differences across the three dataset pairs revealed substantial model-specific asymmetry (Fig. 16). DINOv2-Base showed the smallest mean absolute asymmetry (0.0116), whereas DeiT-Small showed the largest (0.0683). The remaining models lay between 0.0322 and 0.0489. This result matters because a single external test direction can understate generalization risk: a model that transfers well from a broad source to a narrower target may fail more strongly in the reverse direction. The full bidirectional design used by G therefore captures a property that a one-way external-validation table would miss.

4.22. Degradation rate complemented retention area

Area under the retention curve summarizes the complete corruption trajectory, but models with similar areas can degrade at different rates. Figure 17 therefore reports linear retention slopes across severity for each model–corruption

Table 15
Mean absolute directional asymmetry in macro-F1 across the three unordered dataset pairs. Lower is more symmetric.

Model	Mean absolute asymmetry
DINOv2-Base	0.0116
ViT-B/16	0.0322
ResNet50	0.0333
MobileNetV2	0.0380
CLIP ViT-B/32	0.0397
Swin-Tiny	0.0455
DenseNet121	0.0489
DeiT-Small	0.0683

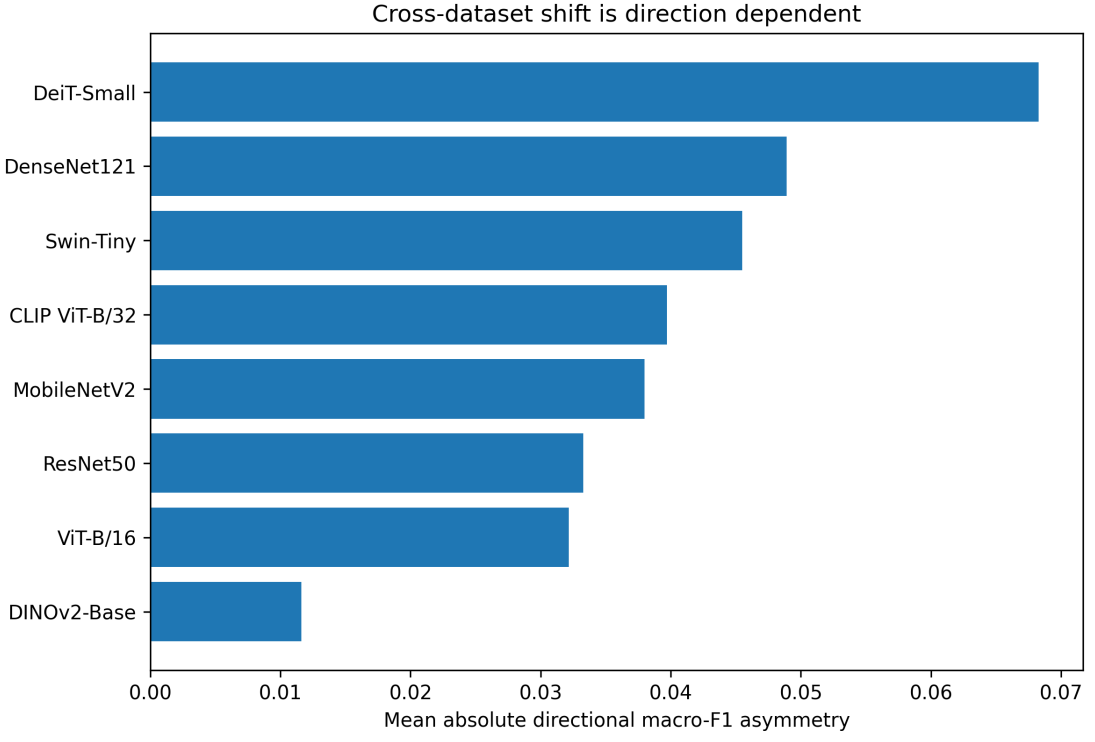


Figure 16: Mean absolute directional asymmetry across bidirectional dataset-pair transfers. Lower values indicate more consistent transfer behavior irrespective of source–target direction.

pair. The analysis confirms that robustness is corruption-specific rather than a single universal property: the same architecture can decline gently under brightness/contrast perturbation yet deteriorate faster under blur or Gaussian noise. Because CRuSS aggregates corruption families only after computing within-family trajectories, a model cannot obtain a strong R merely by excelling on one easy perturbation family.

4.23. Weight-free Pareto analysis supported the leading CRuSS result

The multi-objective dominance analysis provides an aggregation-independent sanity check. DINOv2-Base was non-dominated and, more strongly, Pareto-dominated all seven alternative model families across (A, G, Q, R, L). ViT-B/16 was dominated only by DINOv2-Base and itself dominated CLIP ViT-B/32 and DeiT-Small. ResNet50 was also dominated only by DINOv2-Base and dominated DenseNet121. Other models exhibited trade-offs that prevented a complete ordering. Thus, the leading CRuSS result is not an artifact of equal weights or geometric aggregation: the highest-ranked model is also superior on every constituent axis relative to every comparator in this benchmark.

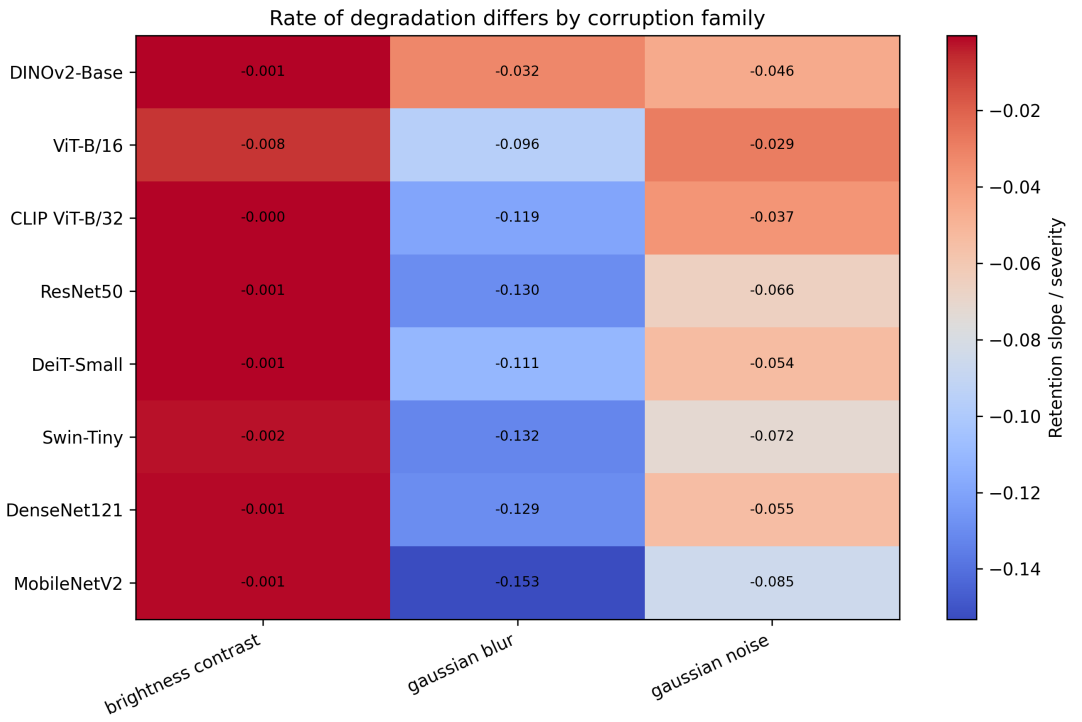


Figure 17: Normalized macro-F1 degradation slope per severity step. More negative values indicate faster loss of performance as corruption strength increases.

Table 16
Pareto-dominance summary across the five CRuSS components.

Model	Models dominating it	Models it dominates
DINOv2-Base	0	7
ViT-B/16	1	2
ResNet50	1	1
Swin-Tiny	1	0
MobileNetV2	1	0
CLIP ViT-B/32	2	1
DenseNet121	2	0
DeiT-Small	3	0

4.24. Interpretive stress tests strengthen the case for a composite reliability view

Taken together, the additional analyses provide three checks against overclaiming. First, low correlation between clean adequacy and the two stress-resilience axes shows that the composite contains genuinely new empirical information rather than merely rescaling accuracy. Second, directional asymmetry demonstrates that external generalization is a relation between a source and target distribution, not an intrinsic scalar attached to a model. Third, Pareto analysis shows that the leading model remains preferred even without committing to a particular aggregation function. Conversely, the middle of the ranking should be interpreted more cautiously: where models trade strengths across components, CRuSS is a transparent summary of those trade-offs rather than proof of universal superiority.

5. Discussion

The central empirical finding is that reliability ranking is not recoverable from clean ranking alone. This is most visible for MobileNetV2: excellent clean macro-F1 did not translate into resilience under progressive corruption or

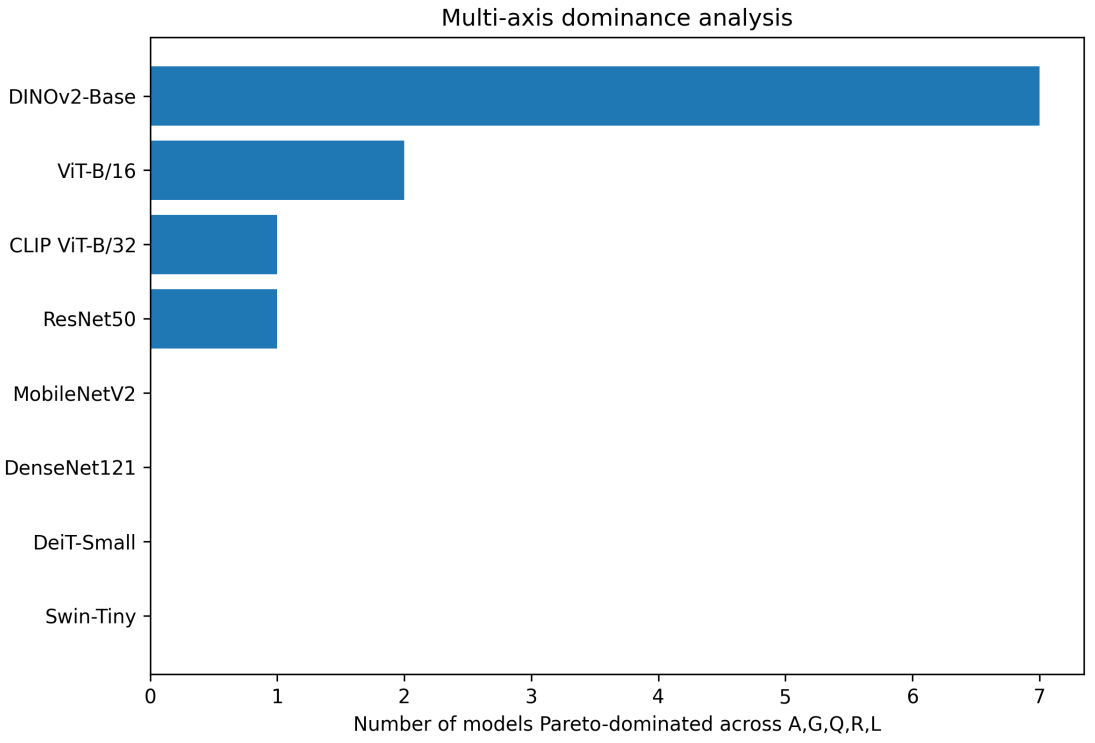


Figure 18: Number of comparator models Pareto-dominated across all five CRuSS dimensions. This analysis does not use aggregation weights.

class-specific perturbation. Conversely, ViT-B/16 and CLIP gained rank once shift behavior was included. These are exactly the cases for which a multi-axis reliability summary is useful: the model-selection question is not "which model has the highest clean F1?" but "which model preserves useful behavior across distinct failure modes?"

CRuSS deliberately differs from a generic weighted score in three ways. First, its components are not arbitrary collections of correlated accuracy metrics; they correspond to distinct deployment questions: clean adequacy, domain retention, probability calibration, global stress resilience, and local-class resilience. Second, the retention components are scale-free relative to the model's own clean baseline, preventing a model from being rewarded merely because the target dataset is easier. Third, geometric aggregation is conjunctive: a severe weakness cannot be fully compensated by a near-perfect component. This design aligns with the view that reliability requires simultaneous satisfaction of multiple conditions rather than maximization of one average metric [1].

The score should nevertheless be used with its component vector. A scalar can simplify model selection, benchmarking, and longitudinal monitoring, but it cannot encode every domain-specific clinical cost. If one error type has a known clinical utility or harm function, that cost should be represented directly in task-specific evaluation rather than hidden inside CRuSS. Likewise, CRuSS does not certify safety, fairness, causal validity, or prospective clinical effectiveness. It is an empirical stress-reliability index.

The provenance analysis is also important. Medical AI papers often describe cross-dataset evaluation as external validation, but public imaging datasets can share upstream sources, augmented descendants, or transformed copies. Our audit found exact duplicates between 10K and 40K and extensive near-duplicate candidates across all three public collections. This limits any claim of patient-level independence. Rather than ignoring the issue, we integrate provenance as an eligibility condition: the score can summarize observed stress behavior, but external-generalization language must reflect what the datasets can support. The PMRAM case demonstrates why exact hashes alone are insufficient when transformations change file bytes while preserving image content.

The strong DINOv2 result is consistent with the hypothesis that large-scale self-supervised representations can improve distributional robustness, but the present work is not designed to establish a causal architectural advantage.

Architecture, pretraining data, optimization, and dataset construction are confounded. The appropriate conclusion is narrower: among the evaluated checkpoints and stress conditions, DINOv2-Base showed the most balanced reliability profile.

5.1. Comparison with conventional evaluation summaries

Table 17 clarifies the intended niche of CRuSS. Conventional metrics remain indispensable because they answer precise questions about correctness, discrimination, ranking, or calibration. A corruption AUC adds progressive stress information, and an external F1 adds cross-dataset information. What they do not provide individually is a bottleneck-sensitive summary spanning all of these deployment dimensions while retaining a component audit trail. CRuSS should therefore be viewed as a second-order summary built on first-order metrics, not as a competitor that makes those metrics unnecessary.

Table 17
Conceptual comparison of common evaluation quantities and CRuSS. A checkmark indicates that the quantity directly represents the property in its standard use.

Quantity	Clean	Cross-domain	Calibration	Stress curve	Local class	Main role
Accuracy / F1	yes	no	no	no	optional	Predictive correctness on one distribution
AUROC / AUPRC	yes	no	no	no	optional	Ranking/discrimination quality
ECE / Brier	no	no	yes	no	no	Probability reliability
External F1	no	yes	no	no	optional	Transfer to one specified target
Corruption AUC	no	no	no	yes	no	Progressive perturbation robustness
Class-local stress	no	no	no	yes	yes	Subgroup/class-specific fragility
CRuSS	yes	yes	yes	yes	yes	Auditable joint shift-reliability summary

This distinction also explains why the score should not be optimized directly as a training objective. Doing so would conflate evaluation with model fitting and encourage benchmark-specific gaming. CRuSS is intended to remain on the evaluation side of the pipeline, with its components selected before final model comparison.

5.2. Interpretation as a measurement model

CRuSS is best understood as a measurement model for an operational construct—shift reliability—rather than as a new classifier metric in the narrow sense. The latent construct cannot be observed directly; instead, it is represented by five measurable behaviors. This framing imposes useful discipline. A component should be included because it represents a distinct failure mechanism, not because adding it improves correlation with a desired ranking. Likewise, a high total score is meaningful only together with the component vector and the documented stress protocol.

The present results provide evidence for three forms of validity. *Discriminant validity* is suggested by the weak association with saturated clean metrics and the large rank inversions. *Convergent validity* is suggested by positive association with independent cross-domain performance. *Sensitivity validity* is suggested by monotonic degradation under progressive corruption and by the ability of the local component to identify class-specific failures. Future work should add predictive validity by testing whether a pre-computed CRuSS score forecasts performance at a genuinely unseen hospital.

5.3. Why the geometric mean is preferable here

The geometric mean is not claimed to be universally superior to arithmetic aggregation. It is appropriate here because the construct is conjunctive: a model with excellent clean accuracy but severe corruption fragility should not receive a near-perfect reliability score. Arithmetic aggregation is more compensatory and would permit a saturated component to offset a weak one. The log-space formulation also makes weighting transparent. If a clinical application

later justifies unequal weights, they can be declared prospectively as exponents whose sum is one, preserving the same mathematical interpretation.

An alternative would be to use a minimum operator, which is even more bottleneck-sensitive. We rejected that choice because it is excessively unstable to one noisy component estimate and discards information from the remaining axes. Harmonic aggregation is another option but has less direct multiplicative interpretation. The geometric mean therefore provides a compromise between strict weakest-link behavior and excessive compensation.

5.4. Relationship to robustness, calibration, and domain adaptation literature

CRuSS evaluates reliability; it does not adapt the model. This distinction separates the work from domain-adaptation methods designed to improve target performance [32]. It also differs from calibration correction methods that modify outputs after detecting acquisition shift [30]. Such interventions could be evaluated with CRuSS before and after adaptation, making the score potentially useful as a standardized outcome for robustness interventions. Likewise, self-supervised or foundation-model approaches that improve transfer can be compared through the component profile rather than only by target accuracy [31].

This positioning is important for biomedical engineering: the contribution is an evaluation instrument, not another architecture. In practice, model developers need both. Adaptation methods aim to reduce shift; reliability metrics determine whether the reduction is broad, calibrated, and class-safe rather than an improvement on one selected endpoint.

5.5. Practical use of CRuSS in a model-development pipeline

A practical deployment workflow can use CRuSS in four stages. During model development, the component vector can reveal whether additional training should target calibration, corruption robustness, or a class-local weakness. During model selection, CRuSS can break ties among models with similarly saturated clean performance. During external validation, the directed retention component can show whether a candidate model transfers consistently across sites rather than only to one favorable target. After deployment, the same conceptual axes can support periodic stress testing when scanners, protocols, or preprocessing pipelines change.

The scalar should not be used as a regulatory pass/fail threshold without task-specific validation. A score of 95 has no universal clinical meaning. Thresholds would need to be tied to predefined operating requirements and validated prospectively. The immediate value of CRuSS is comparative and diagnostic: it provides a transparent summary while preserving the five-dimensional audit trail.

5.6. Implications of the provenance audit

The PMRAM case is methodologically consequential because it demonstrates that evaluation quality and metric quality are inseparable. If the fourth dataset had been retained as independent evidence, high external performance could have been incorrectly interpreted as broad generalization. Exact hashes alone would not have revealed most of the transformed overlap. By treating provenance as an eligibility layer rather than another score component, CRuSS avoids rewarding or penalizing a model for a property of the benchmark itself.

This also affects how the three primary public datasets should be described. They provide useful distributional variation, but incomplete patient metadata and evidence of some cross-dataset duplication mean that the term "independent clinical cohort" would be too strong. We therefore use "cross-dataset" or "external-dataset" evaluation and reserve stronger clinical-generalization claims for future patient-level multi-center validation. This conservative wording strengthens rather than weakens the methodological contribution because it separates what the experiments demonstrate from what they cannot establish.

5.7. What would falsify the usefulness of CRuSS?

A proposed metric becomes more credible when its failure conditions are explicit. CRuSS would add little value in a setting where all five components are nearly perfectly correlated, because the scalar would simply reproduce a conventional ranking. It would also be inappropriate if the task lacks meaningful probability outputs, if no relevant shift/stress environments can be specified, or if one clinically dominant error cost should override all other dimensions. Finally, if future prospective studies show that CRuSS has no association with genuinely unseen-site performance or deployment failures, the metric should not be promoted as a general reliability indicator. These falsifiability conditions provide a concrete agenda for independent validation.

5.8. Minimum reporting standard for CRuSS studies

For reproducible use beyond this benchmark, we recommend that any CRuSS report include both the scalar and the complete component vector; the exact task metric used to define A, G, R, and L; all source and target dataset identities; the directed transfer design; corruption types and numeric severity schedules; calibration definition and binning procedure; the class-local stress target; random seeds; uncertainty procedure; component weights; and a provenance/leakage statement. Reporting only the final scalar would defeat the purpose of an auditable reliability score.

Table 18
Recommended minimum reporting checklist for CRuSS-based evaluation.

Item	Required disclosure
Task metric	Base metric and averaging scheme used for adequacy/retention
Domains	Source/target identity, sample size, and direction of every transfer
Calibration	Definition of ECE (and secondary NLL/Brier where available)
Stress protocol	Corruption family, exact severity values, deterministic/random policy
Local stress	Target class or subgroup and rationale for selection
Aggregation	Component equations, clipping rules, weights, and score scale
Uncertainty	Bootstrap unit, number of replicates, and interpretation of interval
Sensitivity	Weight perturbation and component-ablation results
Provenance	Duplicate/near-duplicate audit and patient-level independence status
Audit trail	Component vector reported alongside the final scalar

5.9. Illustrative model-selection scenarios

The component structure enables different practical decisions without recomputing a hidden metric. If two models have similar CRuSS but one has lower Q, a deployment relying on calibrated probabilities may prefer the other even if the total score is close. If R is weak but G is high, the dominant risk may be acquisition degradation rather than site/domain change, suggesting image-quality monitoring or robustness training. If L is uniquely weak, broad retraining may be less appropriate than class-focused error analysis. Thus the scalar supports ranking, while the components support intervention selection.

The present benchmark illustrates all three cases. MobileNetV2's main weakness is not clean adequacy but global and local stress resilience. DenseNet121 is particularly vulnerable to class-local Gaussian noise. DeiT-Small shows the largest cross-domain calibration degradation. DINOv2-Base is strong across all measured axes and therefore ranks first without requiring a compensatory trade-off. These examples show why the component vector is scientifically more informative than a leaderboard number alone.

5.10. Generalisability of the framework beyond brain MRI

The algebra of CRuSS is task-agnostic, but the operationalization of each axis must remain task-aware. For segmentation, adequacy and retention could use a problem-appropriate overlap or boundary metric selected according to the target structure properties, while calibration may require pixel- or object-level definitions. For detection, A and G could use a declared detection metric and the local component could target a clinically important lesion subtype or size stratum. For physiological signals, the corruption axis could represent sensor noise, lead dropout, or temporal artifacts rather than image perturbations. The framework therefore generalizes through its reliability structure, not through blindly reusing macro-F1 in every domain.

A particularly promising use is comparative evaluation of robustness interventions. Domain adaptation, self-supervised pretraining, calibration correction, and data augmentation can each improve one aspect of reliability while degrading another. A common five-axis outcome could reveal whether an intervention produces broad improvement or merely shifts the failure mode. Such studies would also provide independent evidence about whether the current component set is sufficient or needs extension.

5.11. A practical reliability card for reproducible reporting

To make the framework operational, Table 19 proposes a compact reporting card that should accompany any CRuSS value. The card separates evidence generation from score presentation and makes hidden weaknesses difficult to obscure. In particular, reporting the worst transfer direction and worst stress family guards against an average score being read

without its failure context. The provenance field is equally important: a mathematically valid score computed on a contaminated external set is not valid evidence of external reliability.

Table 19
Recommended reliability card to accompany CRuSS in future studies.

Field	Minimum information to report
Aggregate	CRuSS point estimate and uncertainty interval
Component vector	A, G, Q, R, and L with exact definitions
External validity	All source→target directions, not only the best external result
Calibration	Shifted ECE plus at least one secondary proper-score quantity where available
Stress profile	Corruption families, severity schedule, retention curves, and worst family
Local failure	Pre-declared class or subgroup stress and worst local retention
Stability	Seeds, bootstrap unit, number of replicates, and ranking sensitivity
Provenance	Exact-duplicate and near-duplicate audit outcome for every claimed external set
Decision use	Any prospective minimum component requirement or deployment constraint
Reproducibility	Code version, preprocessing, normalization, checkpoints, and evaluation configuration

The card also clarifies what CRuSS does not claim. It does not convert heterogeneous evidence into a regulatory certification, and it does not imply that a model with a marginally higher score is clinically superior when confidence intervals overlap or when the lower-scoring model satisfies a deployment-specific constraint better. For model development, the aggregate score is most useful for triage; for final deployment, the component vector and worst-case evidence should remain primary.

5.12. Clinical interpretation and responsible use

CRuSS should be used as a screening and comparison instrument, not as a substitute for prospective clinical validation. The score has no direct probabilistic interpretation and does not encode disease prevalence, downstream decision costs, reader interaction, workflow delay, or harm severity. A deployment team could therefore require a minimum class-local resilience or calibration criterion even when aggregate CRuSS is high. This is a feature rather than a limitation of the framework: the component vector remains visible precisely so that a scalar ranking cannot conceal a clinically unacceptable weakness.

A useful reporting pattern is to pair the aggregate score with a "reliability card" containing the five components, bootstrap interval, worst observed source–target transfer, worst corruption family, and worst affected class. Such a card supports model governance because a change in score can be traced to the responsible axis. It also reduces incentives to optimize only the aggregate number: an improvement obtained by sacrificing a critical component remains visible to reviewers and deployment stakeholders.

5.13. Why this work is more than a brain-MRI benchmark

The empirical application uses four-class brain MRI because it provides multiple independently curated datasets and clinically meaningful class-level errors, but the methodological object is the evaluation framework. The same construction can be instantiated for chest radiography, retinal imaging, histopathology, dermatology, ECG image representations, or multimodal diagnostic systems provided that the base task metric and stressors are declared prospectively. In segmentation, for example, A and retention terms could use a task-appropriate overlap or boundary metric; in detection, they could use a pre-declared average-precision formulation. Calibration and stress axes would remain conceptually unchanged. The transferable contribution is therefore the measurement design: adequacy, cross-domain retention, probability reliability, global stress resilience, and local subgroup/class resilience are evaluated jointly while remaining separately auditable.

5.14. Recommended validation hierarchy for future CRuSS studies

Future studies should strengthen evidence in a staged hierarchy. The minimum level is repeated in-distribution testing plus at least one no-adaptation external dataset. A stronger level adds multiple bidirectional external transfers, progressive corruption curves, and class- or subgroup-specific stress. The preferred level further includes patient-level bootstrap uncertainty, prospective or temporally separated external data, pre-specified clinical operating points, and independent replication of the metric ranking. Reporting should distinguish these levels explicitly; a high CRuSS

estimated from a narrow benchmark should not be treated as equivalent evidence to the same score obtained from geographically and temporally independent validation.

5.15. Limitations

Several limitations should guide interpretation. First, all three primary datasets are public, derived, or incompletely documented MRI collections rather than prospectively acquired independent clinical cohorts. Patient IDs and scanner metadata are unavailable, so patient-level leakage and scanner-stratified analysis cannot be completed. Second, global and class-local corruption experiments use one fixed dataset and seed-42 checkpoints; future validation should repeat progressive corruption across source datasets and seeds. Third, the class-specific benchmark targets meningioma only. Extending local stress to all classes would make the local component more comprehensive. Fourth, ECE is binning-dependent and has known limitations; CRuSS therefore treats it as one transparent calibration summary rather than a definitive uncertainty metric. Fifth, equal log-space weighting is a design choice. The Dirichlet sensitivity analysis shows substantial ranking stability, but different applications may justify predeclared domain-specific weights. Finally, the current validation is restricted to four-class brain-MRI classification; demonstrating portability to other modalities, tasks, and prevalence regimes is necessary before claiming broad universality.

5.16. Threats to validity

Construct validity depends on whether the five selected axes adequately represent deployment reliability. They cover several common failure modes but do not include fairness, abstention/selective prediction, interpretability stability, latency, or clinical utility. These omissions are intentional rather than accidental: adding every desirable property would make the score opaque and task-dependent. Independent studies should test whether extensions improve predictive validity without creating redundant components.

Internal validity is strengthened by fixed seeds, deterministic stress generation, no target tuning, and score recomputation from saved result tables. It is limited by the fact that the corruption benchmark uses seed-42 checkpoints on one fixed test set rather than all 120 checkpoints. Consequently, R and L capture model-family stress behavior under a controlled representative setting, not a full factorial estimate across every source and seed.

External validity is the principal limitation. Public brain-MRI datasets provide substantial scale but imperfect provenance and limited scanner/site metadata. The results therefore support a methodological claim about metric behavior more strongly than a clinical claim about hospital deployment. Prospective multi-center validation is required before interpreting CRuSS as a clinically validated reliability measure.

Statistical conclusion validity is constrained by only eight model families at the model-ranking level. Bootstrap intervals exploit environment-level repetition, but model-level correlation tests remain low-powered. We therefore emphasize effect patterns, rank inversions, and sensitivity analyses rather than dichotomous significance testing. This separation of empirical effect from inferential certainty is especially important for metric-validation studies.

5.17. Implications for future work

The next validation step should test CRuSS prospectively on independently acquired multi-center cohorts with explicit patient identifiers and scanner/site metadata. A second priority is outcome-based construct validation: the score computed on development stress tests should predict future performance on a genuinely unseen site. Third, CRuSS can be generalized to segmentation or detection by replacing macro-F1 with task-appropriate baseline metrics while preserving the retention/calibration/stress structure, consistent with problem-aware metric selection [1, 2]. Finally, future work can study whether component thresholds, rather than only the total score, provide useful deployment gates.

6. Conclusion

We introduced CRuSS, a transparent, bounded reliability score for medical image classifiers evaluated under distributional shift and stress. Across eight model families, three MRI datasets, 120 clean runs, 240 cross-domain evaluations, controlled corruptions, and class-specific perturbations, CRuSS exposed substantial rank changes hidden by near-saturated clean metrics. DINOv2-Base achieved the most balanced profile, whereas MobileNetV2 demonstrated that high clean performance can coexist with marked stress fragility. Weight sensitivity and environment bootstrap supported the stability of the main ranking. Equally important, a provenance audit disqualified an apparently strong fourth external dataset from independent-generalization claims, underscoring that reliability evaluation depends on both metrics and data integrity. CRuSS is best used as an auditable complement to conventional task metrics: the scalar score summarizes cross-shift reliability, while its components identify why a model succeeds or fails.

Ethics statement

This study used previously released public image datasets and did not involve new participant recruitment, intervention, or access to identifiable private health records. The authors did not attempt to infer patient identities. The datasets were accessed and analyzed under their respective public-access terms and licenses.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All image datasets used in this study are publicly available. The 7K brain-MRI collection (7,023 images) is available from Kaggle at https://www.kaggle.com/datasets/rm1000/brain-tumor-mri-scans. The 10K collection (10,417 images) is available from Mendeley Data, version 2, at https://data.mendeley.com/datasets/x69nzw9zzw/2 (DOI: 10.17632/x69nzw9zzw.2). The 40K collection (40,100 images) is available from Kaggle at https://www.kaggle.com/datasets/lamyaafahim/denoised-mri-brain-tumor-dataset-40k. The fourth operationally unseen PMRAM source is available from Mendeley Data, version 1, at https://data.mendeley.com/datasets/m7w55sw88b/1 (DOI: 10.17632/m7w55sw88b.1). As described in the provenance audit, PMRAM was not treated as an independent external-generalization cohort after substantial transformed-image overlap was identified. The derived clean-evaluation tables, cross-domain prediction summaries, corruption summaries, class-specific stress summaries, CRuSS component tables, sensitivity analyses, and provenance-audit summaries used to generate the reported results are included in the accompanying submission package.

CRediT authorship contribution statement

Moinul Hossain: Conceptualization, Data Curation, Software, Formal Analysis, Visualization, Writing – Original Draft Preparation. **Sabah Nushra:** Data Curation, Investigation, Software, Validation **Md Kishor Morol:** Methodology, Software, Validation, Writing – Review & Editing. **Liew Tze Hui:** Methodology, Supervision, Validation, Writing – Review & Editing.

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